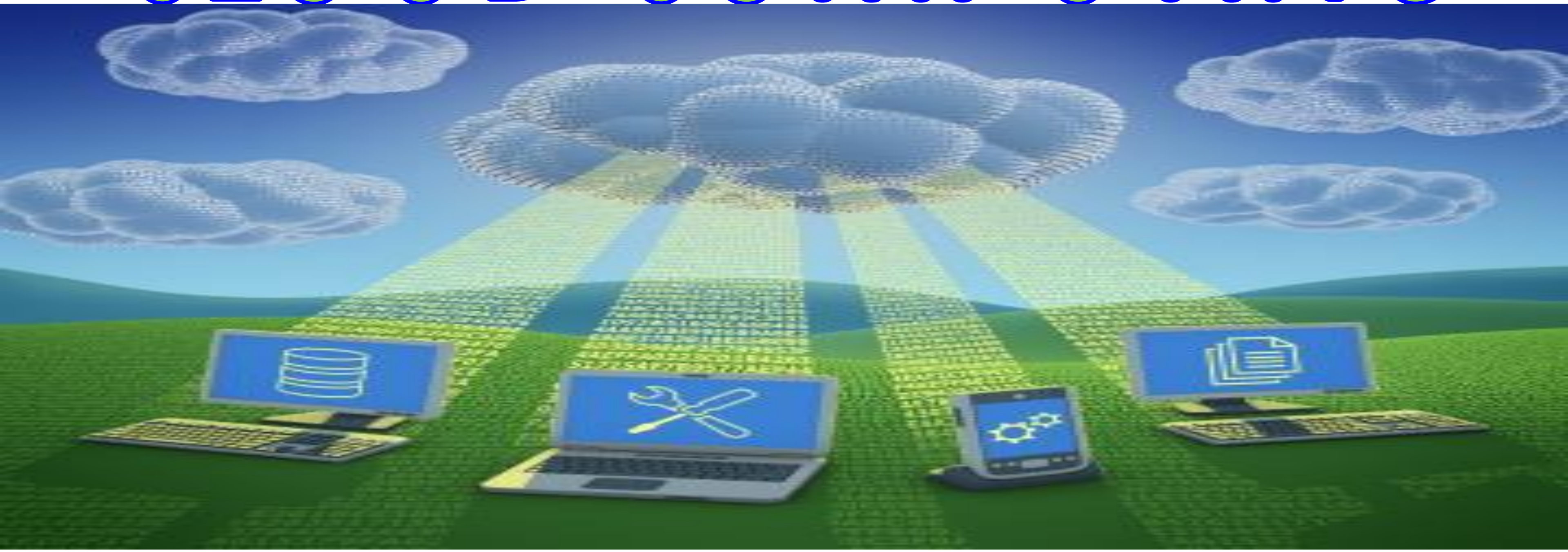
CLOUD COMPUTING



UNIT-I

WHAT IS THE CLOUD?

- The term cloud has been used historically as a something else for the Internet.
- This concept dates back as early as 1961, Professor John McCarthy suggested that computer time-sharing technology might lead to a future where computing power and even specific applications might be sold through a utility-type business model.
- This idea became very popular in the late 1960s, but by the mid-1970s the idea faded away when it became clear that the IT-related technologies.

- It was during this time of revitalization (something) that the term cloud computing began to emerge in technology circles.
- Cloud computing refers to the delivery of computing resources over the Internet.
- Instead of keeping data on your own hard drive or updating applications for your needs, you use a service over the Internet, at another location, to store your information or use its applications.

- Cloud services allow individuals and businesses to use software and hardware that are managed by third parties at remote locations.

- Examples of cloud services include online file storage, social networking sites, webmail, and online business applications.

❖ introduction to Cloud computing

- "Cloud Computing" to put it simply, means "Internet Computing." The Internet is commonly visualized as clouds; hence the term "cloud computing" for computation done through the Internet.

- Cloud computing provides the facility to access shared resources and common infrastructure, offering services on demand over the network to perform operations that meet changing business needs.

- The location of physical resources and devices being accessed are typically not known to the end user.
- It also provides facilities for users to develop, deploy and manage their applications 'on the cloud', which entails virtualization of resources that maintains and manages itself.

❖Definitions

- Cloud computing – "a model for enabling ubiquitous (appearing), convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

❖Definitions

- Cloud computing is using the internet to access someone else's software running on someone else's hardware in someone else's data center. -Lewis Cunningh

- According to Wikipedia:

- "Cloud computing is Internet-based computing, whereby shared resource, software, and information are provided to computers and other devices on demand, like the electricity grid.“

❖ Why cloud services are popular?

Cloud Computing

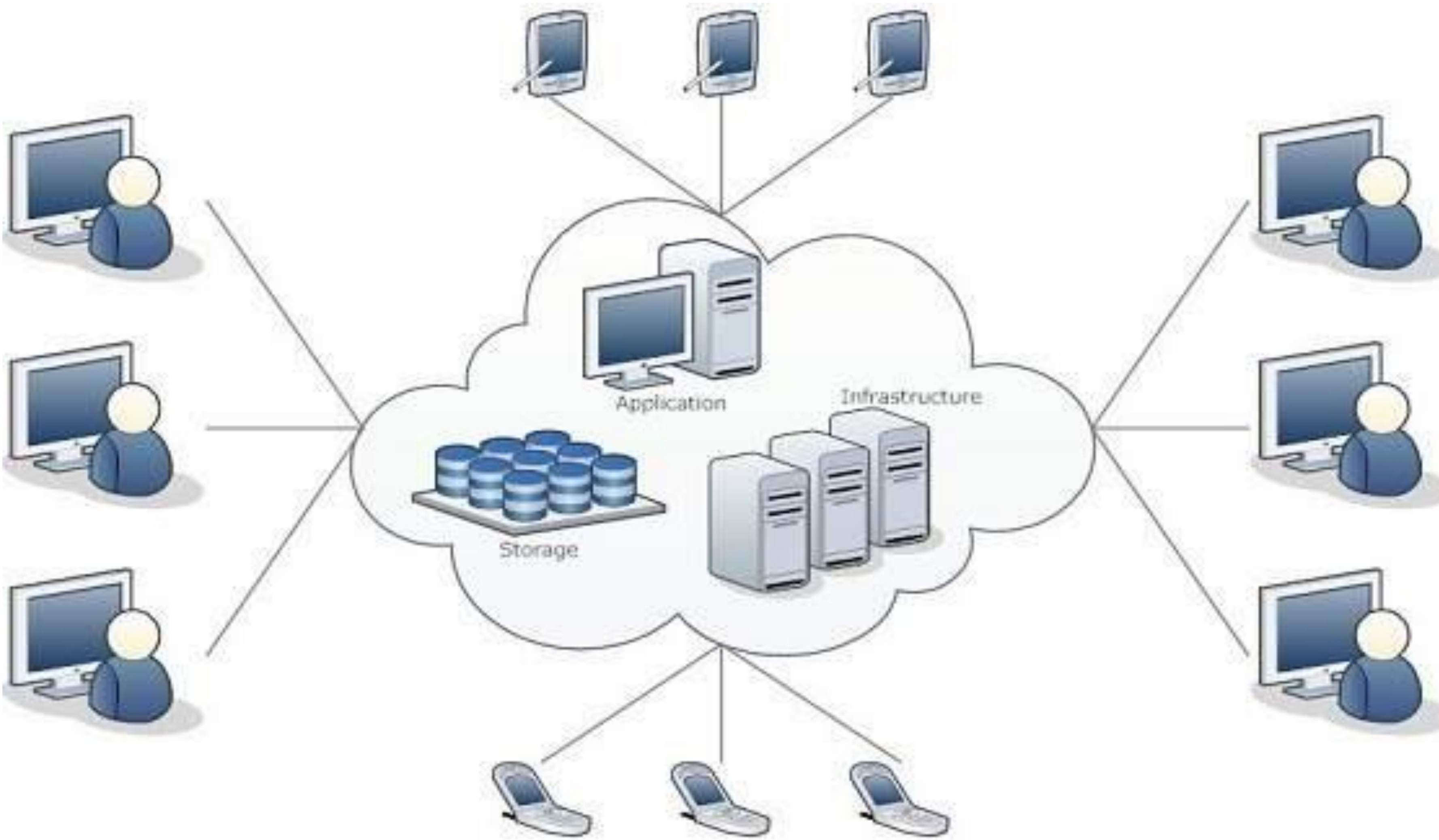


Fig : Cloud Computing Architecture

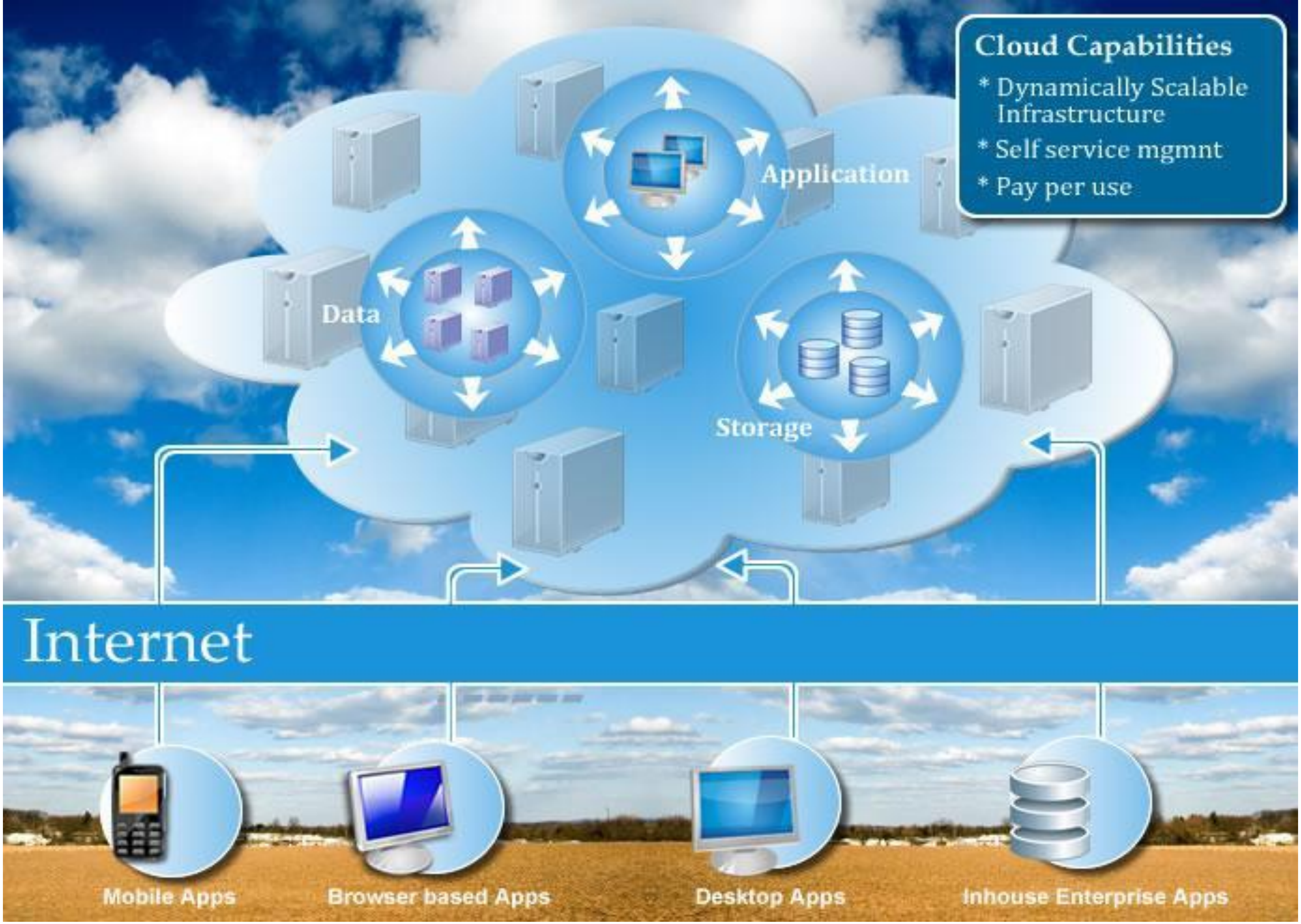
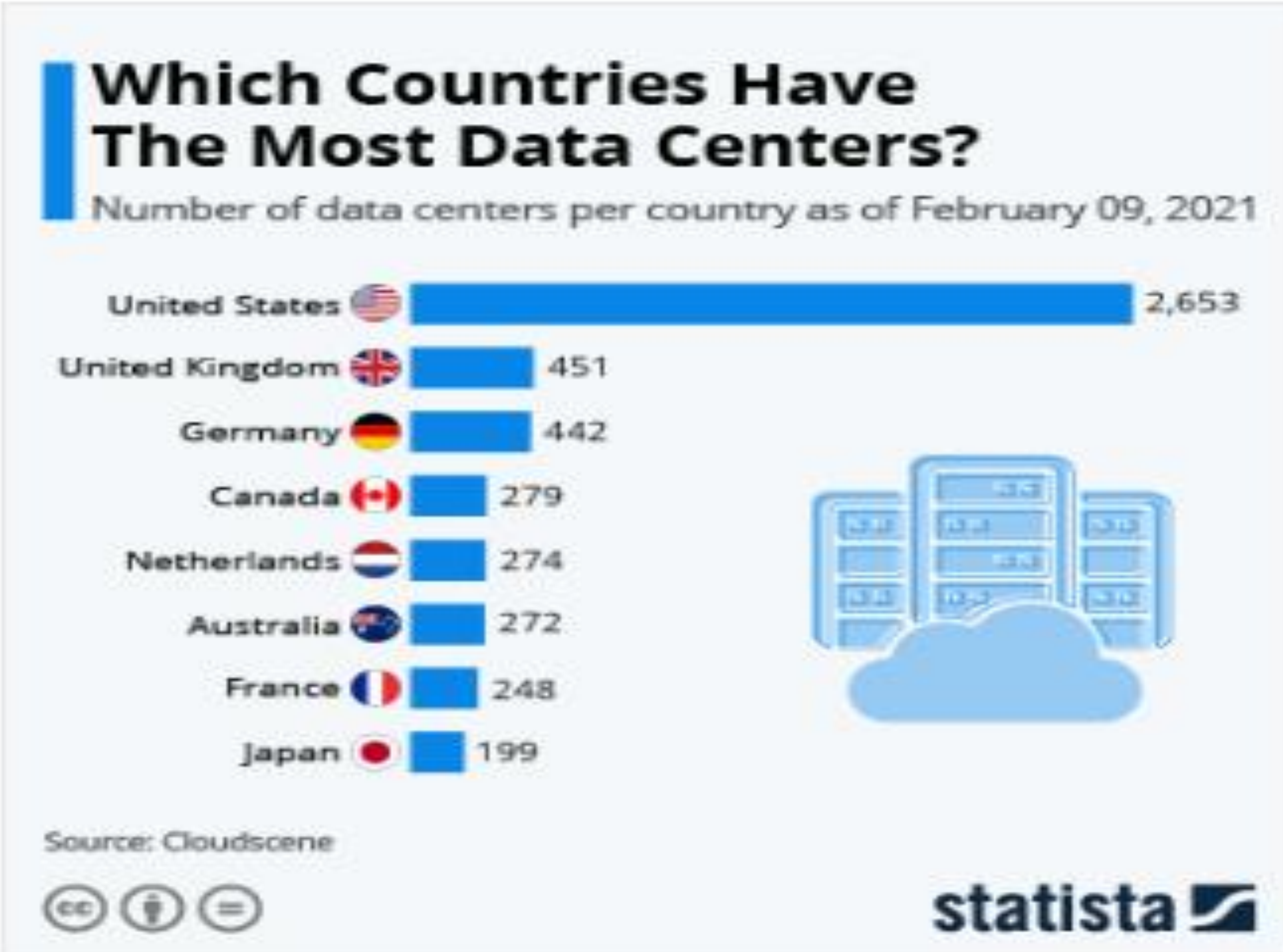
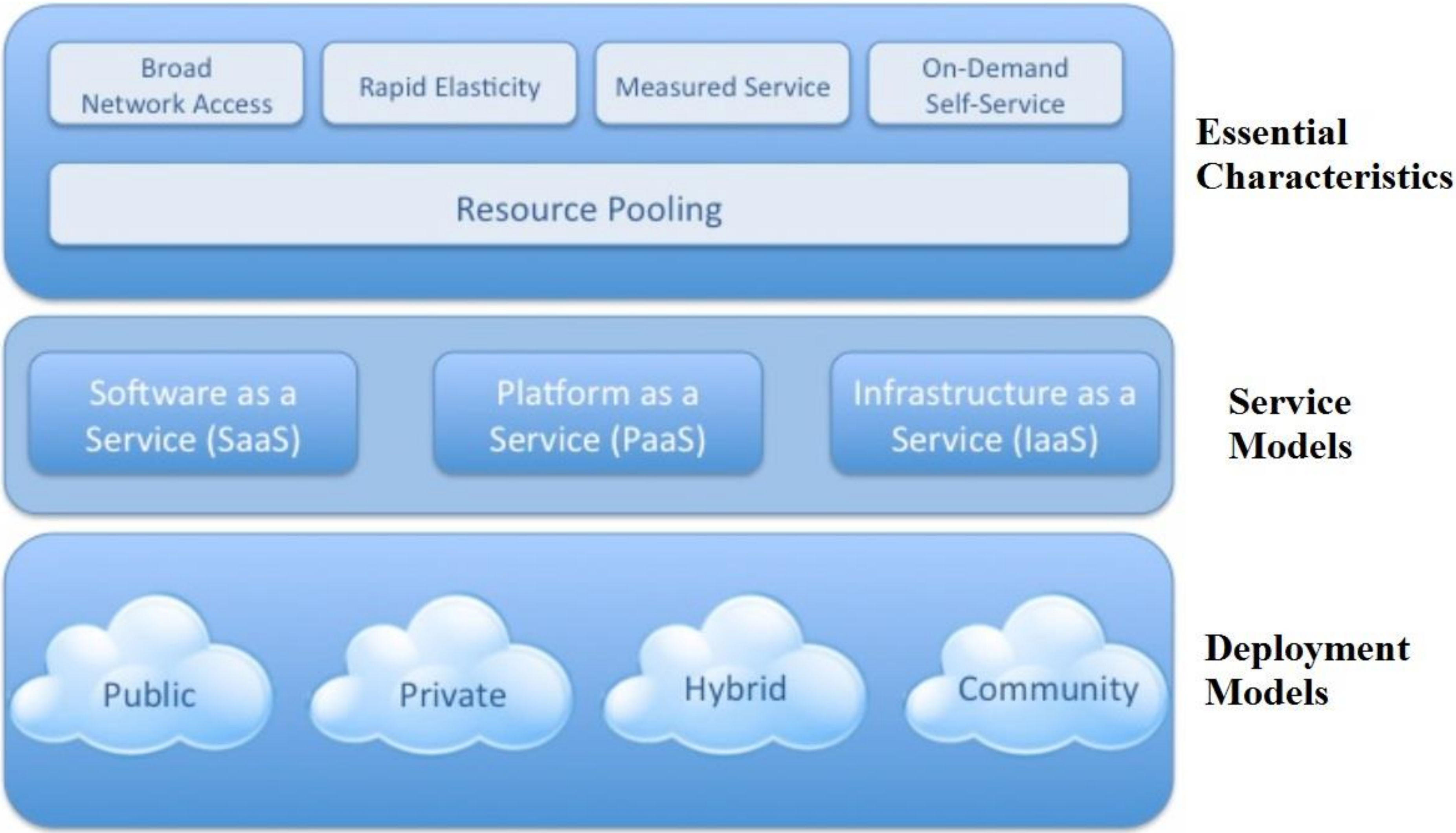


Figure : Conceptual view of cloud computing

INFOGRAPHICS ON THE TOPIC



❖ Cloud Computing Architecture



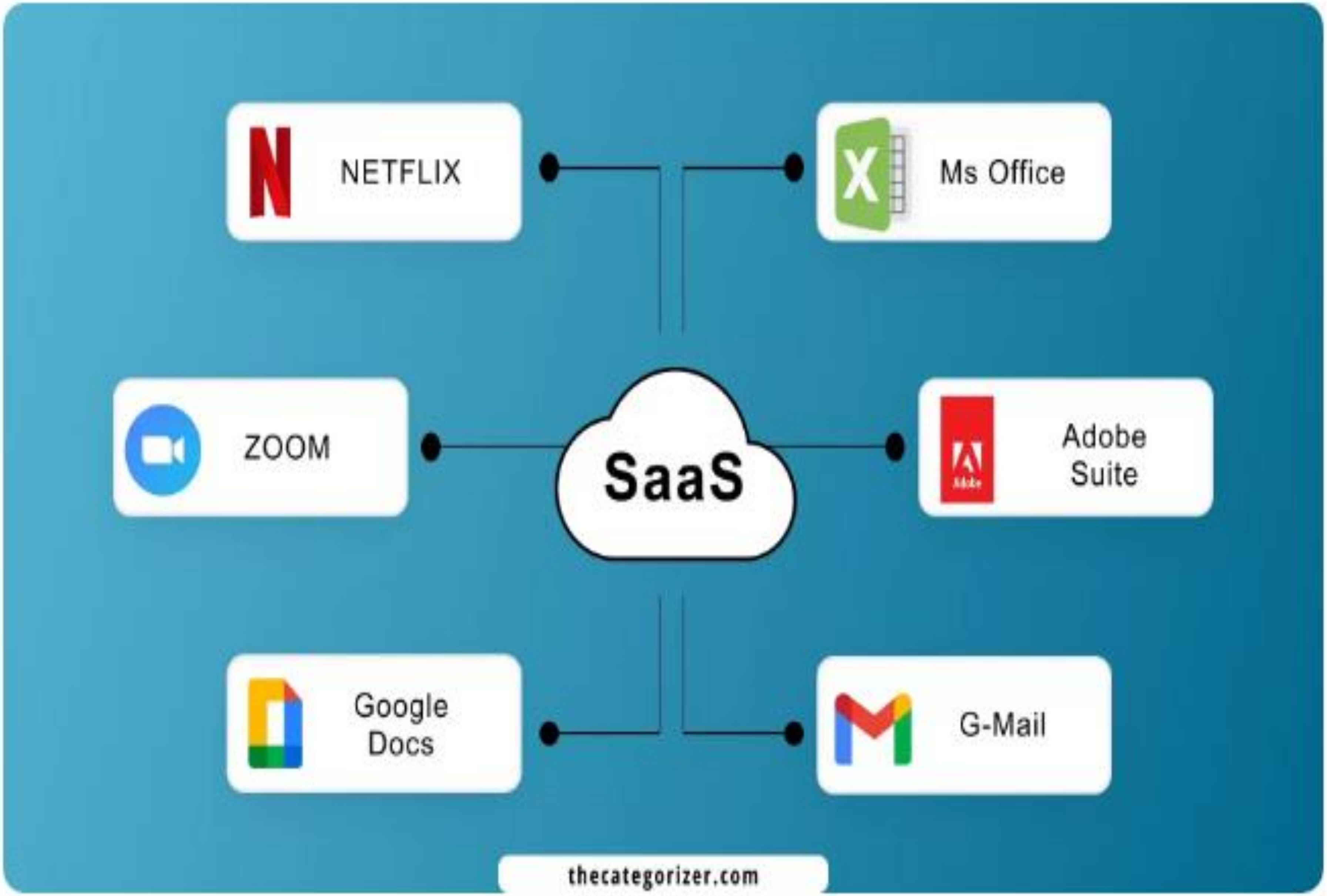
NIST Visual Model of Cloud Computing Definition

❖Essential Characteristics

- **Broad network access -** Broad network access allows services to be offered over the **Internet or private networks**.
- **Rapid elasticity -**Capabilities can be rapidly and elastically provisioned - in some cases automatically to quickly **scale out**, and **scale in**.
- **Measured service-** Use of a service is measured and customers are billed accordingly.

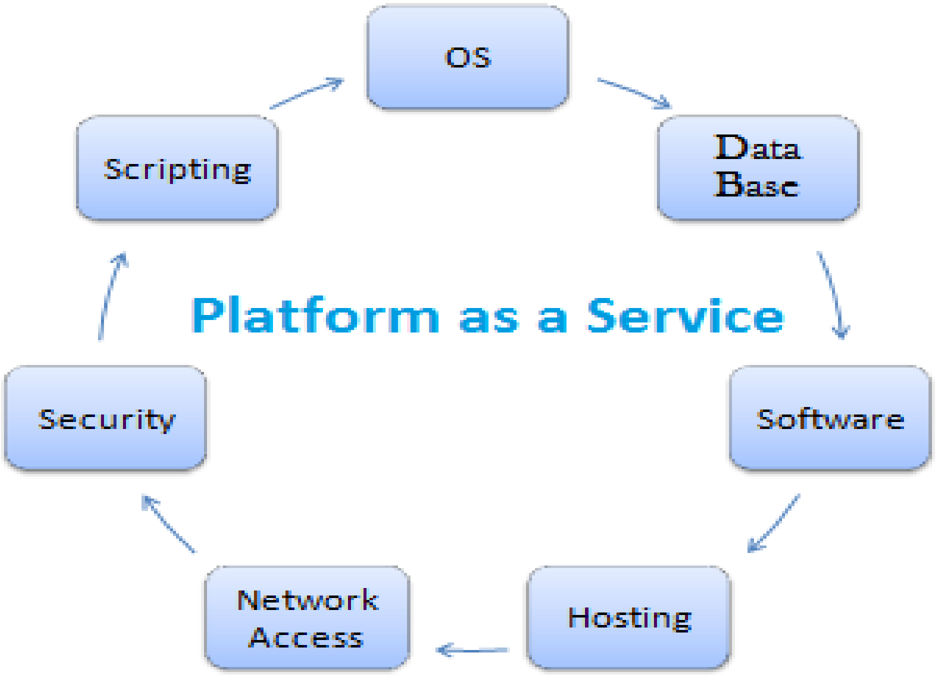
- **On-demand self-service-** On-demand self service means that customers **(usually organizations)** can request and manage their own computing resources.
- **Resource pooling**

  -- Resources can be dynamically assigned and reassigned according to customer demand.


NETFLIX
Ms Office
ZOOM
SaaS
Adobe
Suite
Google
Docs
G-Mail
thecategorizer.com

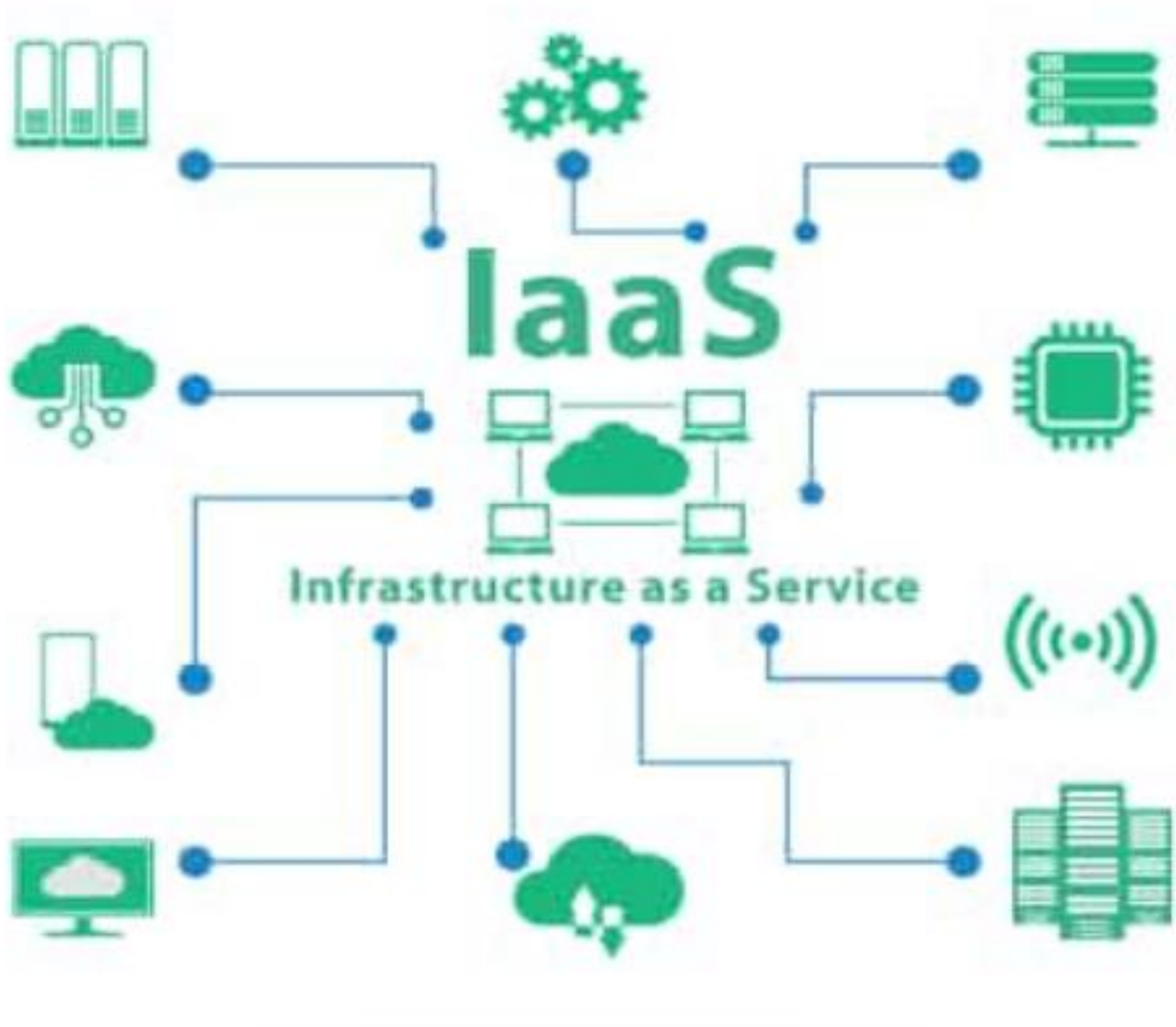
Where PaaS works?

- All the all technical stack requirements are met by platform offerings.
- Automation is paramount in testing, build, release.



Advantages of PaaS cloud Computing Layer

- Lower risk
- Simplified Development
- Scalability


IaaS
Infrastructure as a Service

➤Cloud Service Models

Platform Type	Common Examples
SaaS	Google Workspace, Dropbox, Salesforce, Cisco WebEx, Concur, GoToMeeting
PaaS	AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, OpenShift
IaaS	DigitalOcean, Linode, Rackspace, Amazon Web Services (AWS), Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE)

➢Cloud Service Models

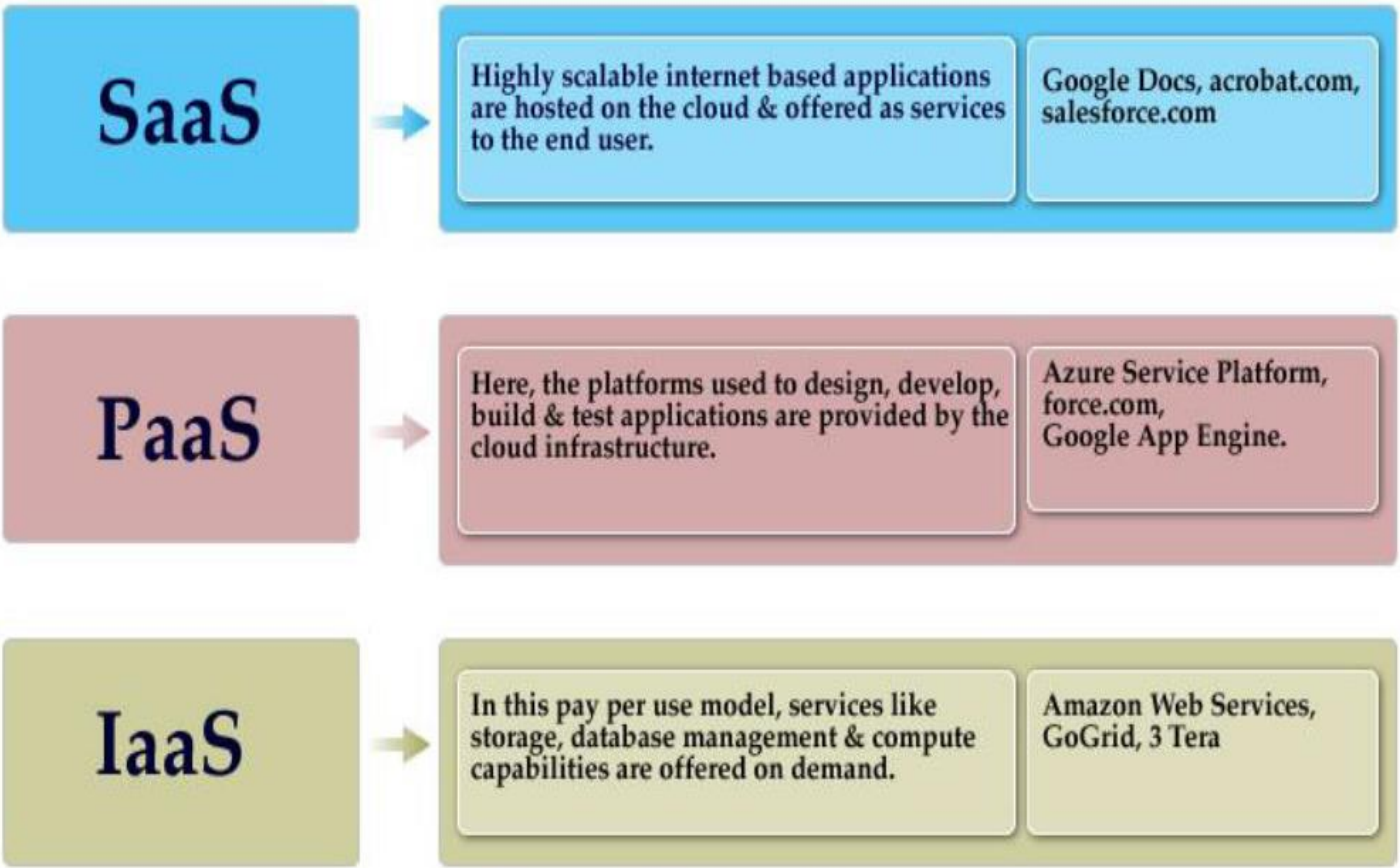
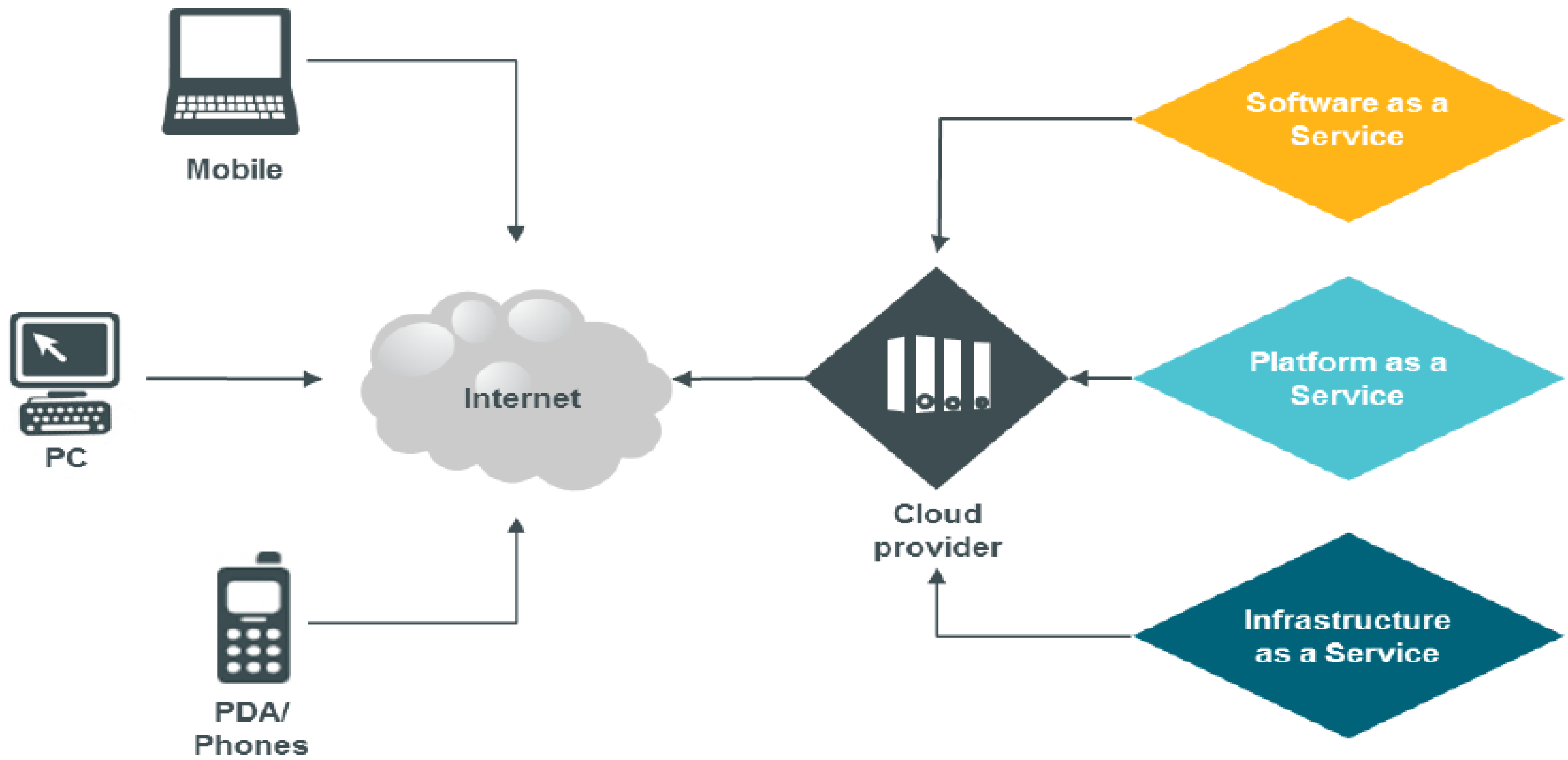


Figure 2: Cloud models

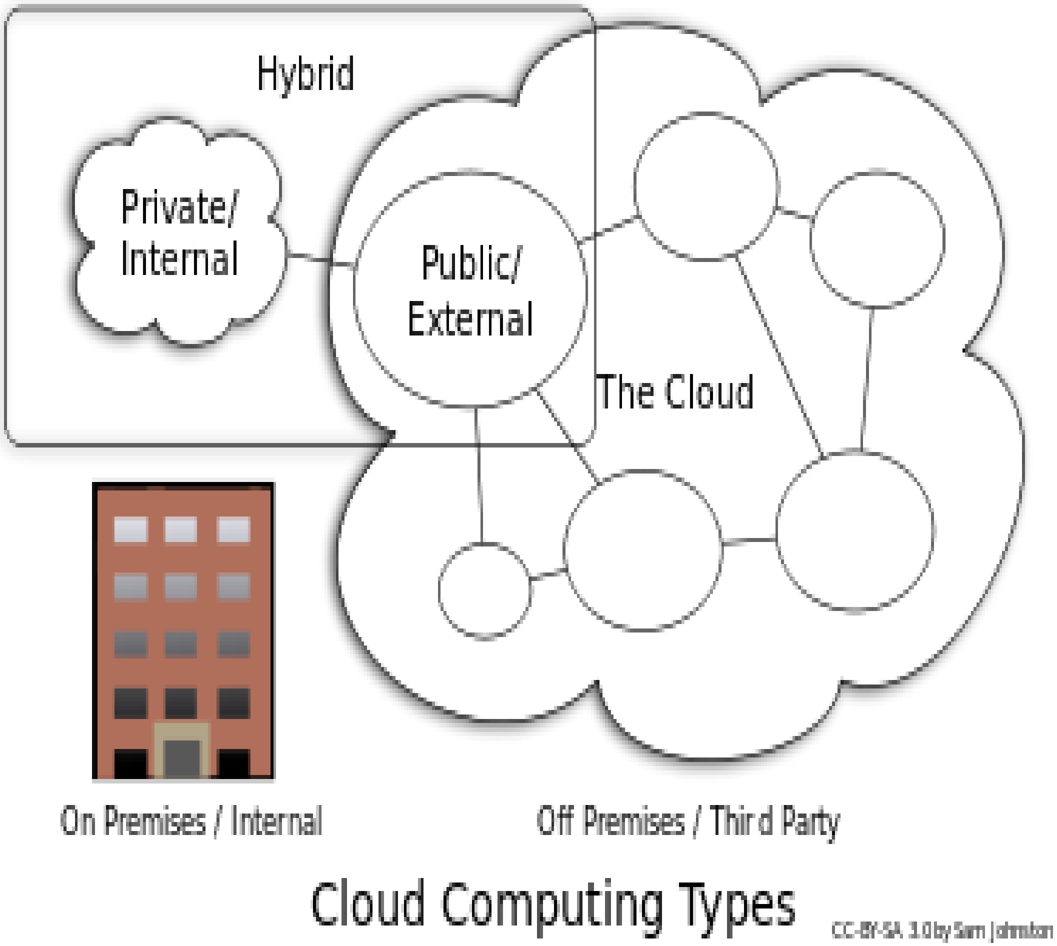
❖ Cloud Service Models

- **Software as a Service (SaaS)**
- **Platform as a Service (PaaS)**
- **Infrastructure as a Service (IaaS)**



❖Cloud Deployment Models

- **Private Cloud**
- **Public Cloud**
- **Hybrid Cloud**
- **Community Cloud**



❖Advantages

- Cost saving: You pay for what you use
- Easy on installation and maintenance
- Increased storage
- Highly automated
- Flexibility
- Better mobility
- Shared resources
- Back up and restoration

❖Disadvantages

- Data security and privacy
- Network connectivity and bandwidth
- Service unavailability due to power outage
- Dependence on outside agencies
- Limited flexibility
- Long term stability of service provider

Cloud Computing

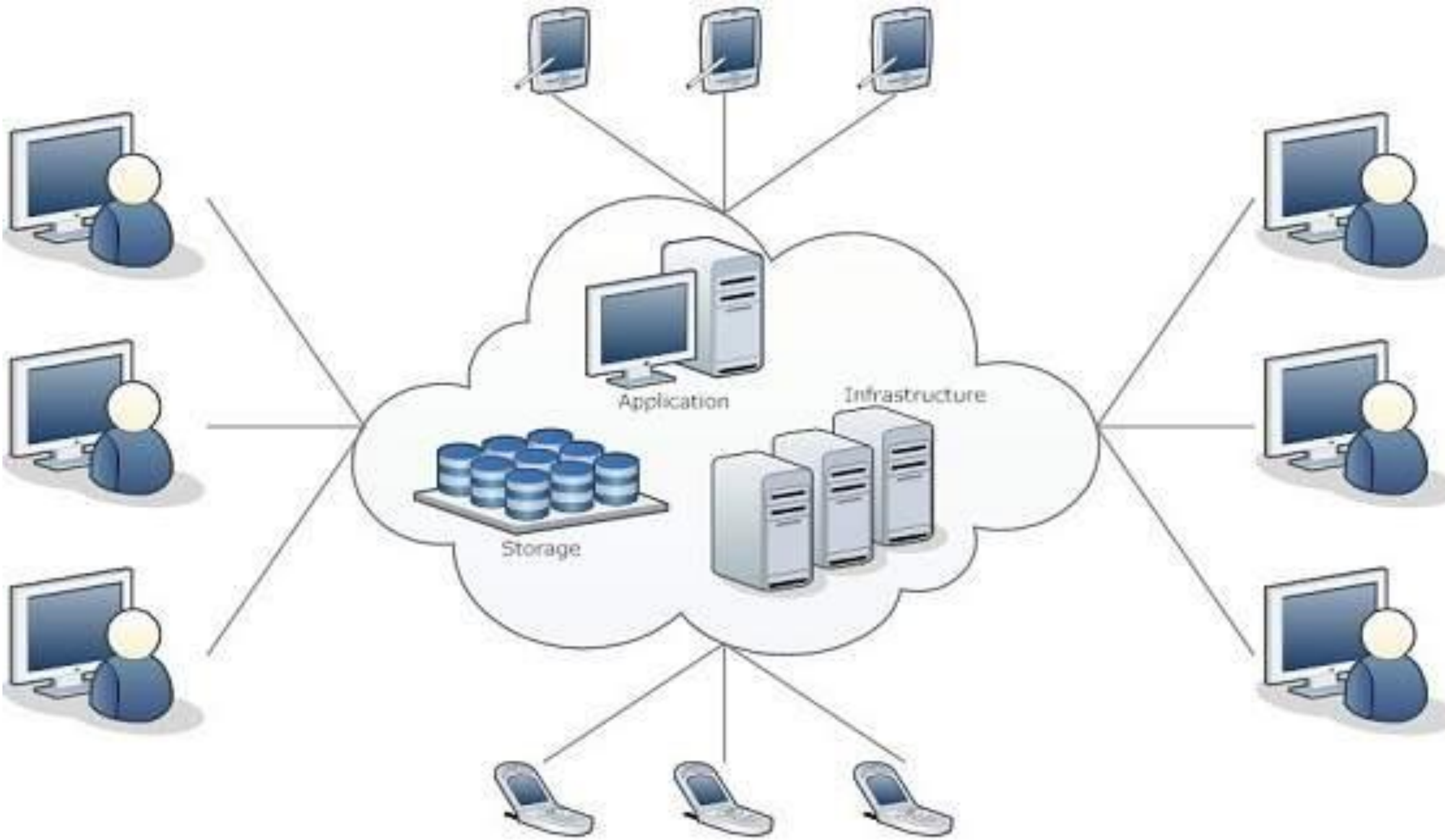


Fig : Cloud Computing Architecture

Characteristics of Cloud Computing

- **On-demand self-service.**
  - A consumer can unilaterally provision computing capabilities such as server time and network storage as needed automatically, without requiring human interaction with a service provider.
- **Broad network access.**
  - Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs) as well as other traditional or cloud based software services.

Characteristics of Cloud Computing

- **Resource pooling.**
  - The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- **Rapid elasticity.**
  - Capabilities can be rapidly and elastically provisioned. Cloud resources can be rapidly scaled up or down based on demand.
  - There are two types of scaling options :

1. Horizontal scaling(scaling out): Horizontal scaling or scale-out involves launching and provisioning(supply) server resources.
2. Vertical Scaling(scaling up) : Vertical scaling or scaling up involves changing the computing capacity assigned to the server resources while keeping the number of server resources constant.

Characteristics of Cloud Computing

- **Measured service.**
  - Cloud systems automatically control and optimize resource usage by leveraging a metering capability at some level of abstraction appropriate to the type of service.
  - Resource usage can be monitored, controlled, and reported - providing transparency for both the provider and consumer of the service.
- **Performance**
  - Cloud Computing provides improved performance for applications

    since the resources available to the applications can be scaled up or down based on the dynamic application workloads.
- **Reduced Costs**
  - Cloud computing provides cost benefits for applications as only as much computing and storage resources as required can be provisioned dynamically, and upfront investment in purchase of computing assets to cover worst case requirements is avoid.

Characteristics of Cloud Computing

- **Outsourced Management**
  - Cloud Computing allows the users to out source the IT infrastructure requirements to external cloud providers.
  - The outsourced nature of the cloud services provides a reduction in the IT infrastructure management costs.
- **Reliability**
  - Applications deployed in cloud computing environments generally have a higher reliability since the underlying IT infrastructure is professionally managed by the cloud service.
  - Cloud service providers specify and guarantee the reliability of cloud in the form of (**Service Level Agreement)** SLA's.
  - Most cloud providers promise 99.99% uptime guarantee for the cloud resources, which may often be expensive to achieve with in-house IT infrastructure.

Characteristics of Cloud Computing

- **Multi-tenancy**

— The multi-tenanted approach of the cloud allows multiple users to make use of the same shared resources.

— Modern applications like e-commerce, Business-to-Business, Banking and Financial, Retail and Social Networking applications that are deployed in cloud computing environment are multi-tenanted applications.

— There are two forms of Multi-tenancy

1. **Virtual Multi-tenancy** : In this Computing and storage resources are shared among multiple users. Multiple tenants are served from virtual machines that execute concurrently on top of the same computing and storage resources.
2. **Organic Multi-tenancy** : In organic multitenancy every component in the system architecture is shared among multiple tenants, including hardware.

Cloud Models

Cloud Service Models

- NIST categorized into three basic service models which are -
  - Infrastructure-as–a-Service (IaaS)
  - Platform-as-a-Service (PaaS)
  - Software-as-a-Service (SaaS)

Infrastructure-as–a-Service (IaaS)

- It is the most basic level of service.
- **IaaS** provides access to fundamental resources such as physical machines, virtual machines, virtual storage, etc.

  Ex : Amazon s3 (Amazon Simple Storage Service)

Platform-as-a-service (Paas)

- PaaS provides the runtime environment for applications, development and deployment tools, etc.

  Ex : Google App Engine .

Cloud Models

Software-as-a-service (Saas)

- **SaaS** model allows to use software applications as a service to end-users. Ex : Facebook ,twiter etc .

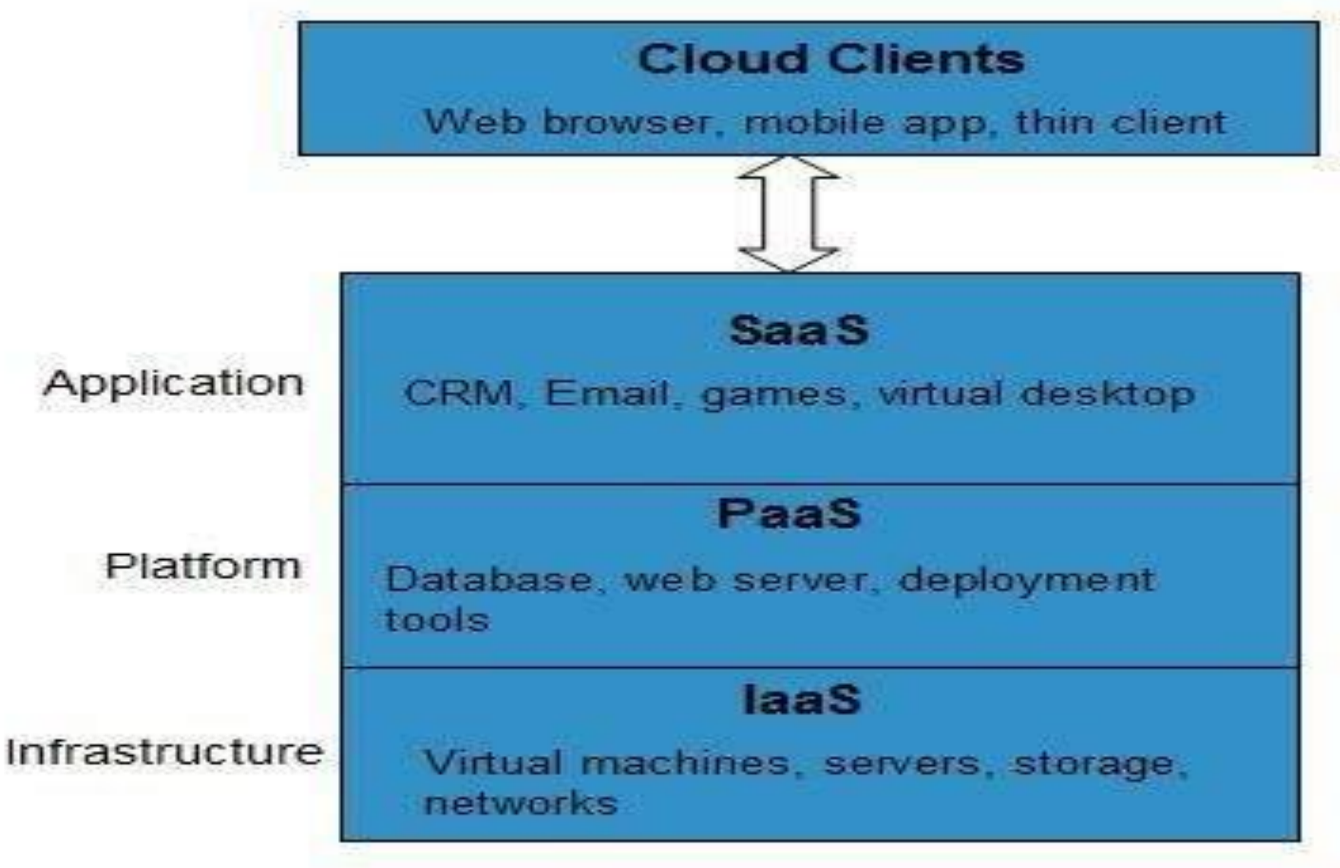


Fig : Cloud Computing Service Models

Cloud Models

- NIST also defines four cloud deployment models as follows :

1. Public Cloud
2. Private Cloud
3. Hybrid Cloud
4. Community Cloud

1. Public Cloud

- In this cloud service are available to the general public or a large group of companies .
- Cloud resources are shared among different users.
- The cloud services are provided by a third party cloud provider.
- It suits for users who want to use cloud infrastructure for development and testing of applications and host applications in the cloud to serve large workloads, without upfront investment in IT infrastructure.

Cloud Models

2. Private Cloud

- In the private cloud deployment model , cloud infrastructure is operated for exclusive us of a single organization.
- Cloud infrastructure can be setup on premise or off-premise and may be managed internally or by a third-party.
- Private clouds are best suited for applications where security is very important and organizations that want to have very tight control over their data.

Cloud Models

3. Hybrid Cloud

- It combines the features of both public and private clouds.
- The individual cloud retain their unique identities but are bound by standardized or proprietary technology that enables data and application portability.
- Hybrid clouds are best suited for organizations that want to take advantage of secured application and data hosting on a private cloud and at the same time cost saving by hosting applications and data in public clouds.

Cloud Models

4. Community Cloud

- In this the cloud services are shared by several organizations that have the same policy and compliance considerations.
- These are best suited for organizations that want access to the same applications and data and want the cloud costs to be shared with the large group.

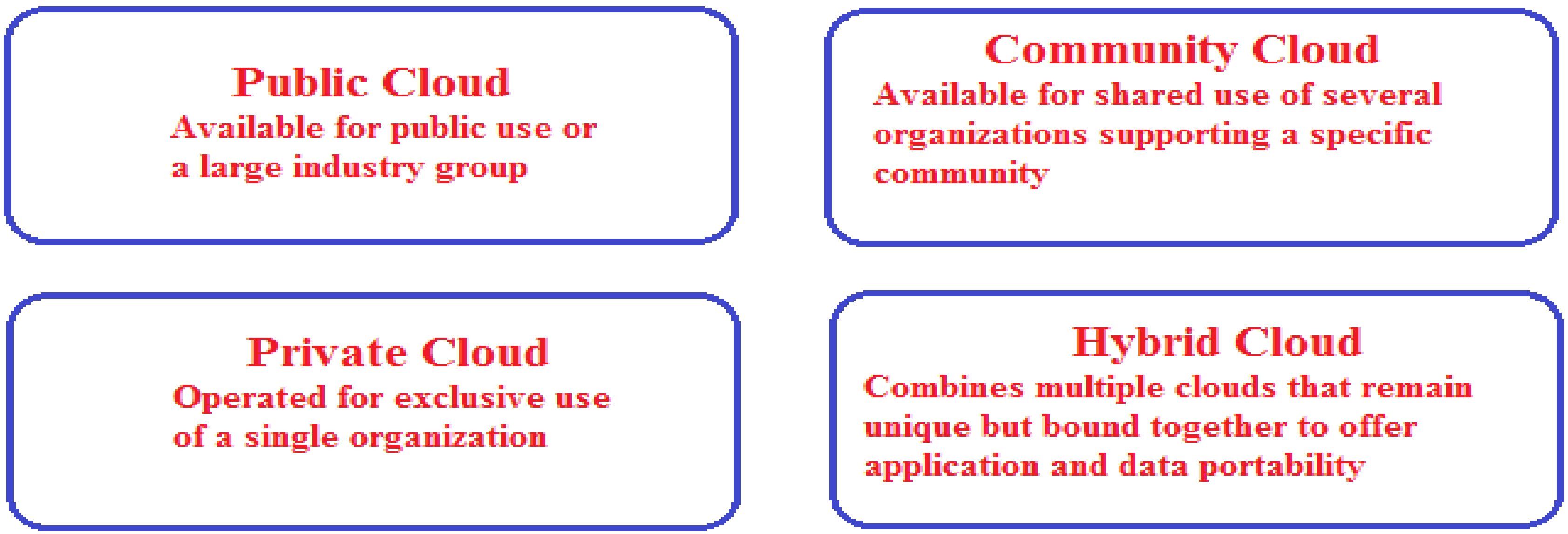


Fig : Cloud Deployment models

Cloud Services Examples

IaaS :

Amazon EC2

- Amazon Elastic Compute Cloud (EC2) is an Infrastructure as a Service offering from amazon.
- It is a web service that provides computing capacity in the form of virtual machines that are launched in Amazon cloud environment.
- It allows users to launch instances on demand using a simple web-based interface.
- Amazon provides pre-configured Amazon Machine Images(AMI) which are templates of cloud instances.
- Users can also create their own AMIs with custom Applications , Libraries and data.
- Users can load their applications on running instances and rapidly and easily increase or decrease capacity to meet the dynamic application performance requirements.

Cloud Services Examples

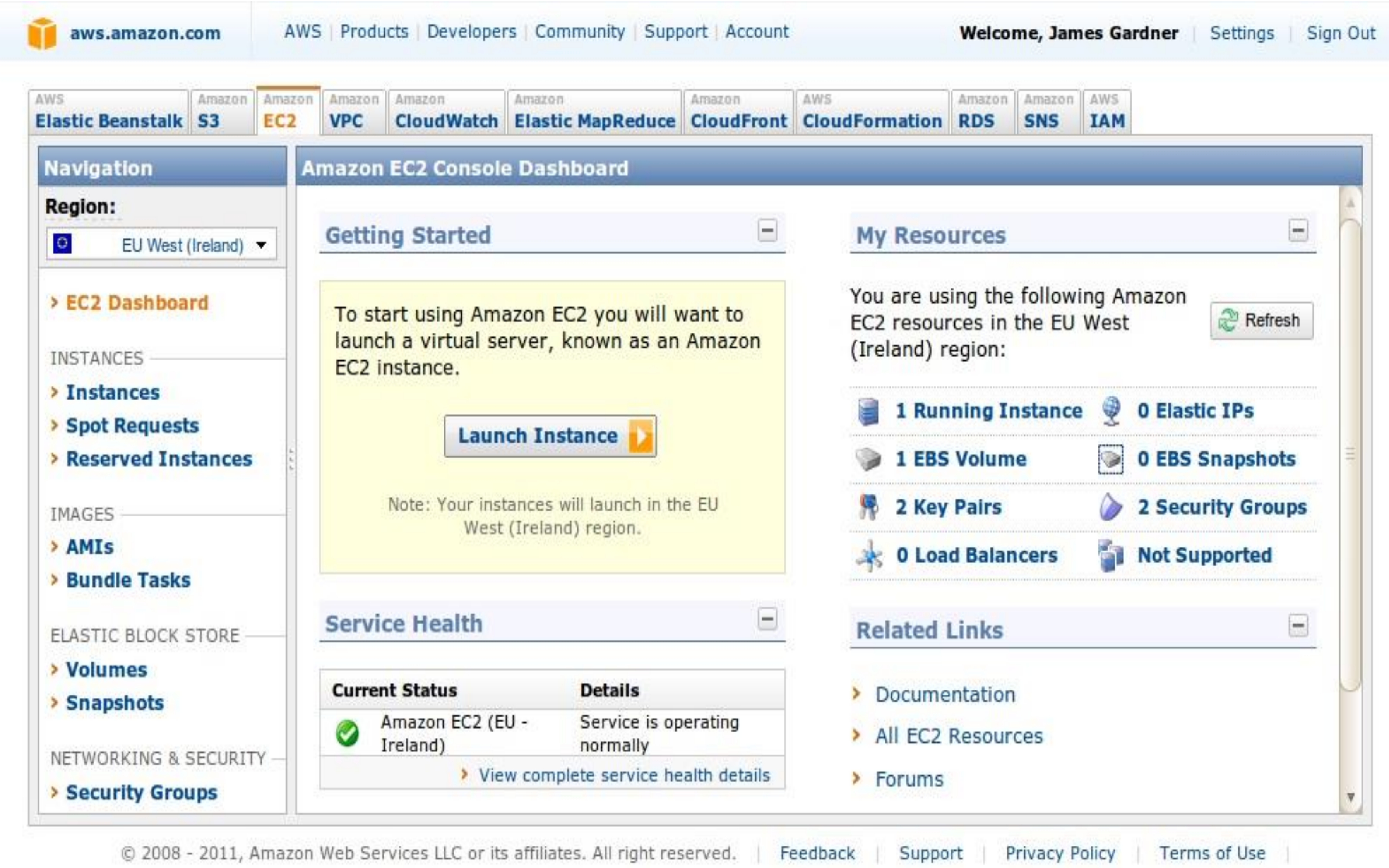
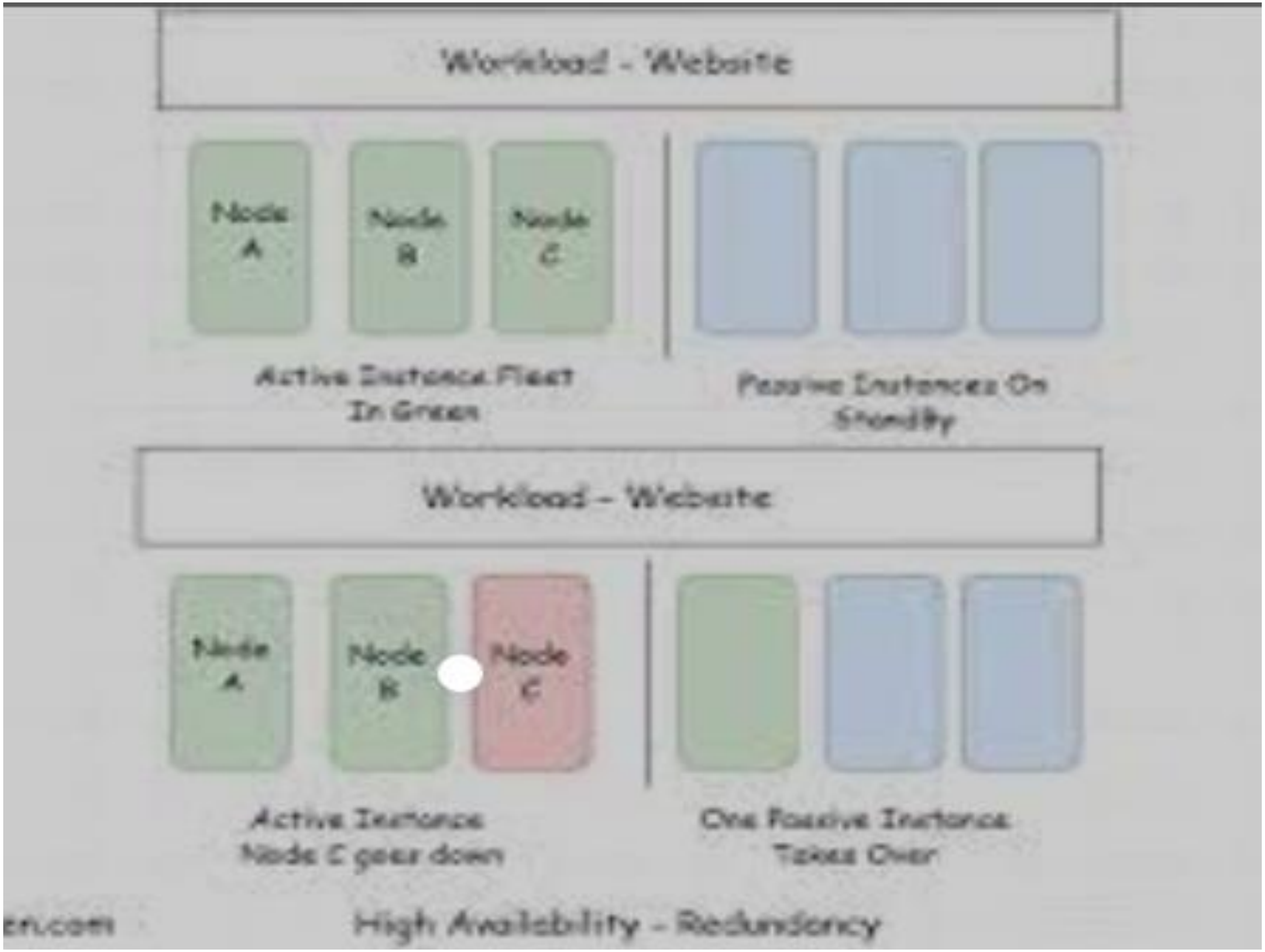


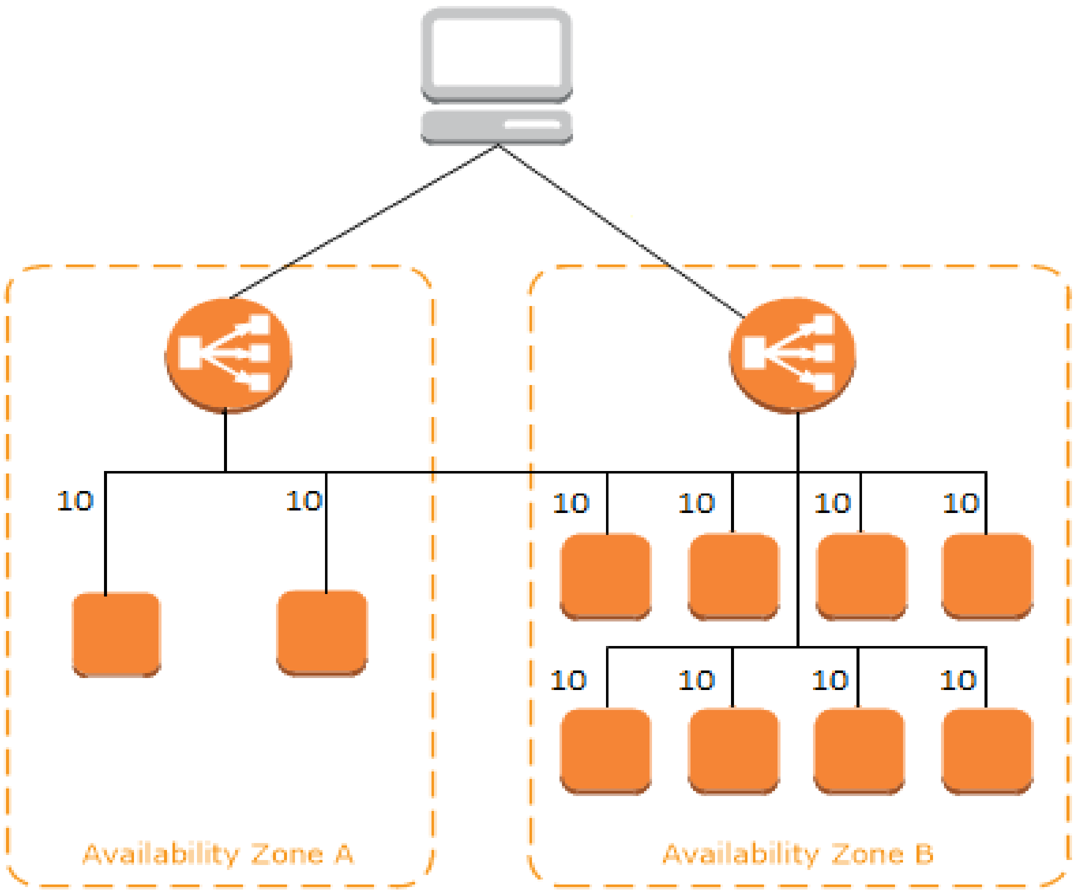
Fig : Amazon EC2 Dashboard

Instances


Workload - Website
Node A
Node B
Node C
Active Instance Fleet In Green
Passive Instances On Standby
Workload - Website
Node A
Node B
Node C
Active Instance Node C goes down
One Passive Instance Takes Over
High Availability - Redundancy

Cloud Services Examples

- Amazon EC2 provides a number of features for building scalable and reliable applications such as auto scaling and elastic load balancing.



Cloud Services Examples

Google Compute Engine

- Google Compute Engine (GCE) is an **IaaS** offering from google.
- GCE provides virtual machines of various computing capacities ranging from small instances to high memory machine types.

Cloud Services Examples

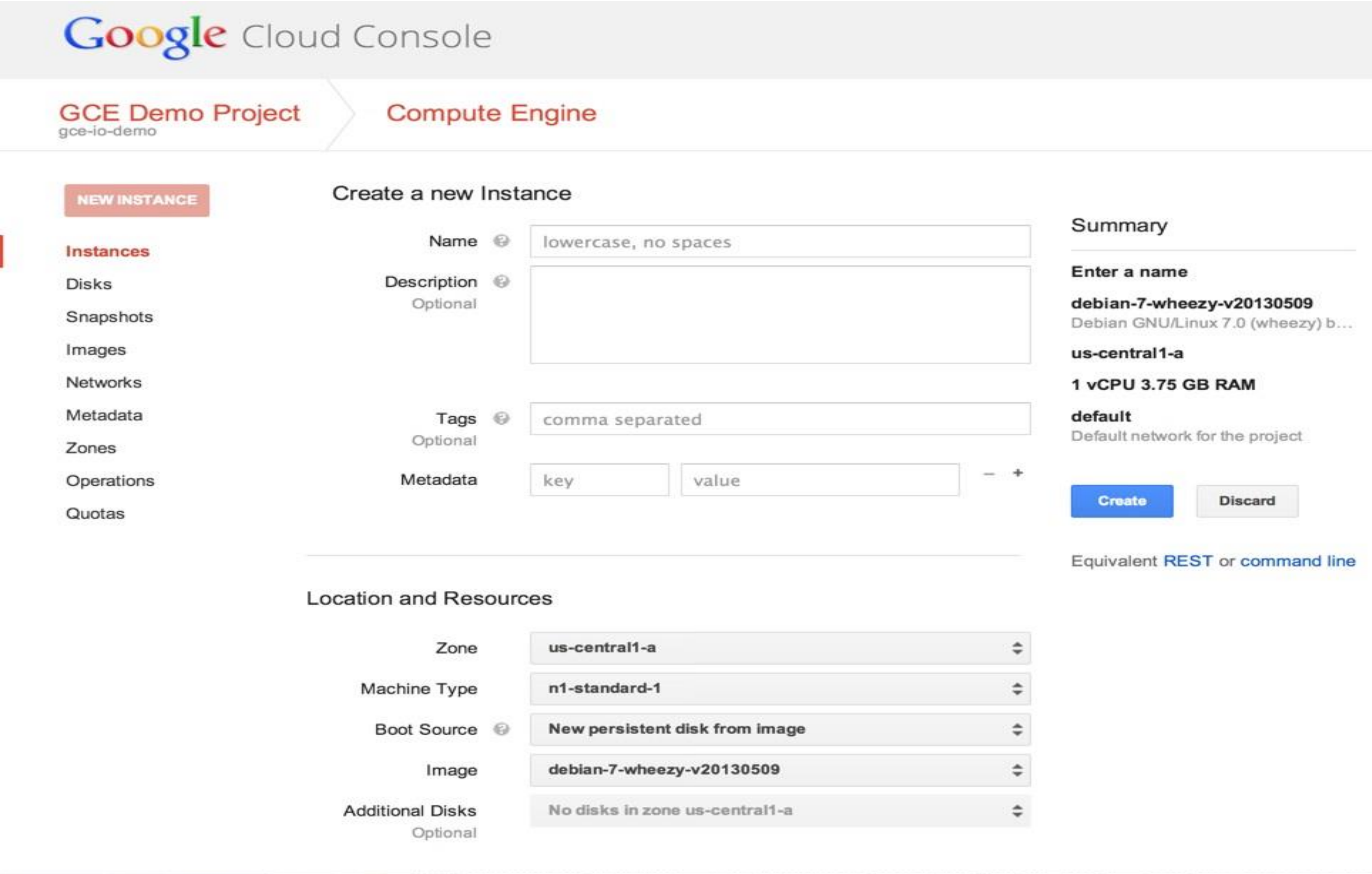


Fig : Google Compute Engine Dashboard

Windows Azure VM’s

- It is an **IaaS** offering from microsoft.
- Azure VMs provides virtual machines of various computing capabilities from small instances to memory intensive machine types.

Cloud Services Examples

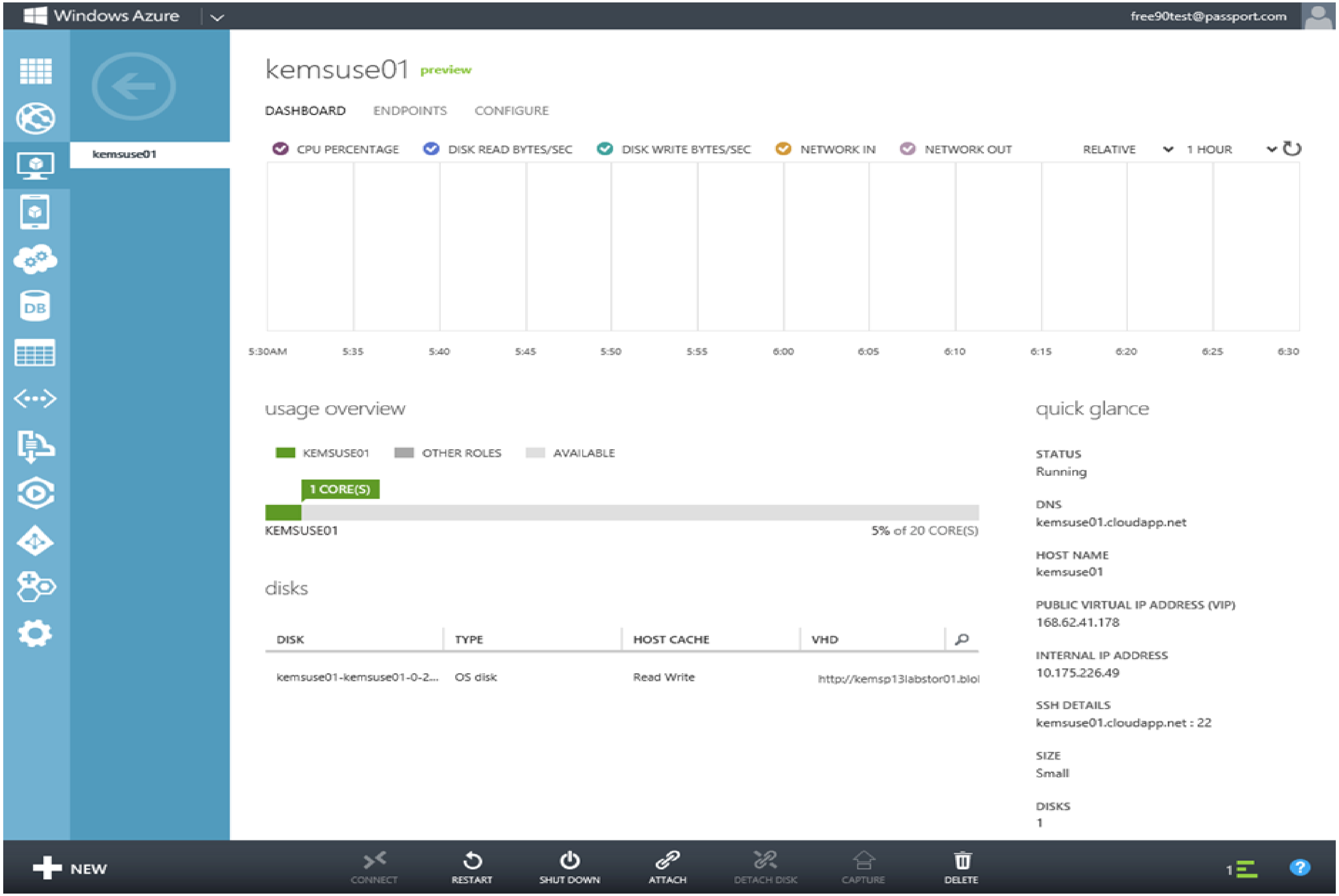


Fig : Windows Azure VM dashboard

Cloud Services Examples

PaaS:

Google App engine(GAE)

- It is a platform-as-a-service offering from google.
- GAE is a cloud-based web service for hosting web applications and storing data.
- GAE allows users to build scalable and reliable(quality) applications that run on the same system that power Google's own applications.
- GAE provides sdk for developing web applications software that can be deployed on GAE.
- GAE supports applications written in several programming languages **like java and python**.
- The pricing model for GAE is based on the amount of computing resources used.

Cloud Services Examples

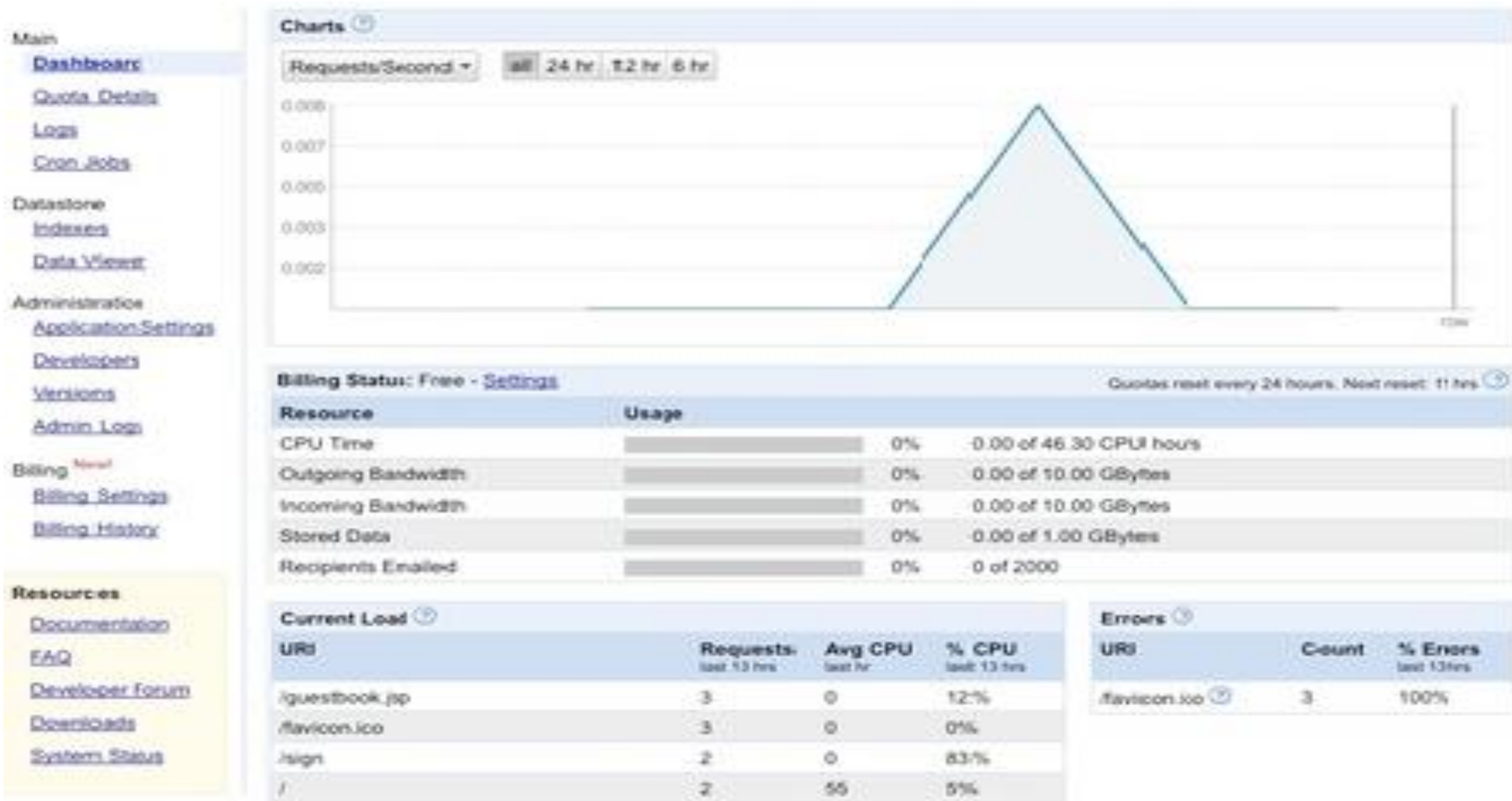


Fig : Google App Engine

Cloud Services Examples

SaaS :

Sales force

- Sales force is a cloud based customer relationship management SaaS offering.
- Users can access CRM(**Customer Relationship Management**) application from anywhere through internet-enabled devices such as **workstations , laptops, tablets and smart phones.**
- It allows sales representatives to manages customer profiles, track opportunities, optimize campaigns from lead to close and monitor the impact of campaigns.

Cloud Services Examples

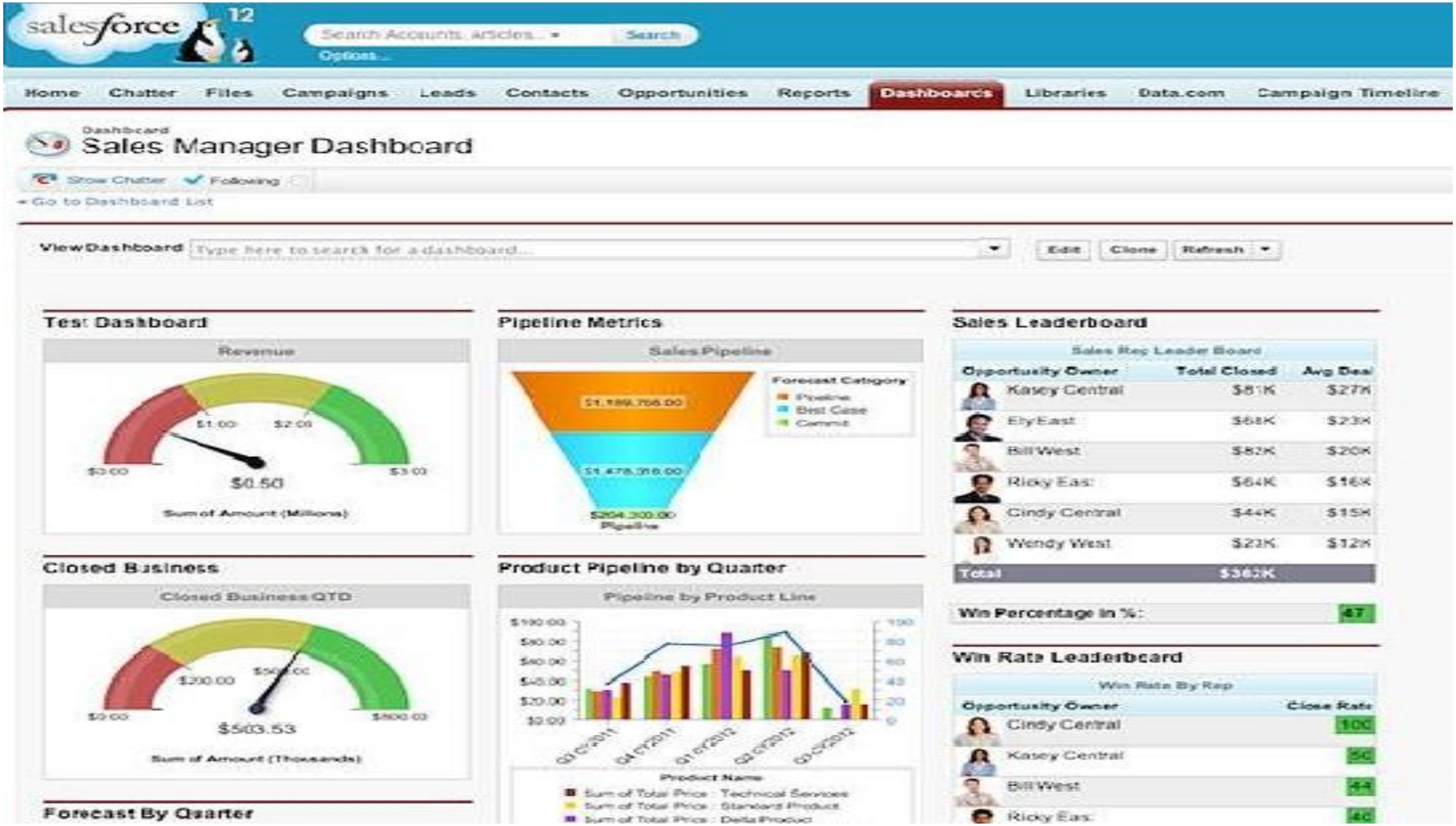


Fig : Sales force dash board

Cloud Computing

Benefits of Cloud Model

- **Reduce spending on technology infrastructure :** Maintain easy access to your information with minimal upfront spending. Pay as you go **(weekly, quarterly or yearly),** based on demand.
- **Globalize your workforce on the cheap :** People worldwide can access the cloud, provided they have an Internet connection.
- **Streamline processes :** Get more work done in less time with less people.
- **Reduce capital costs :** There's no need to spend big money on hardware, software or licensing fees.
- **Improve accessibility.** You have access anytime, anywhere, making your life so much easier!
- **Monitor projects more effectively.** Stay within budget and ahead of completion cycle times.

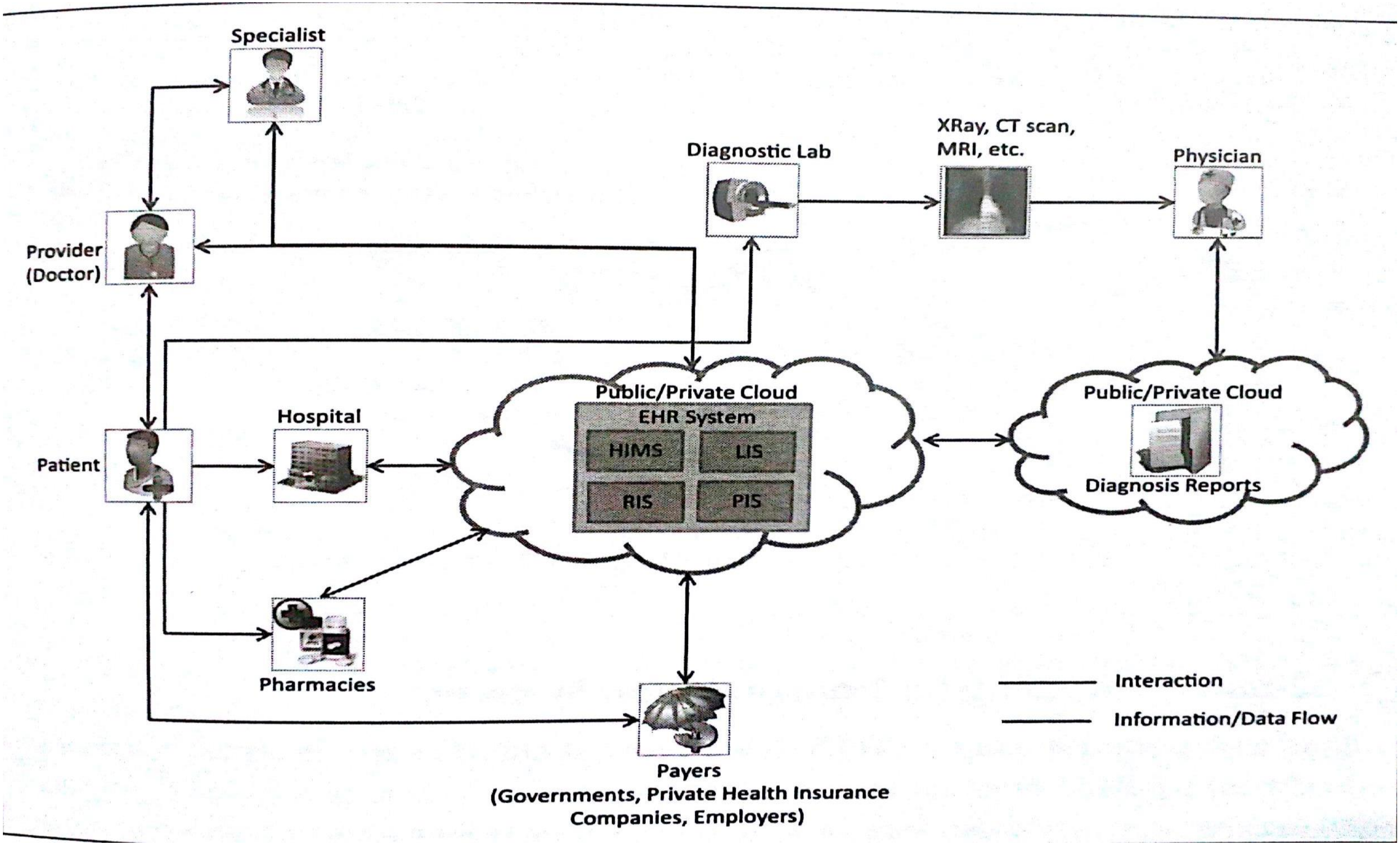
- **Minimize licensing new software :** Stretch and grow without the need to buy expensive software licenses or programs.
- **Improve flexibility :** You can change direction without serious “people” or “financial” issues at stake.

Limitations of Cloud Model

- Large model sizes
- Specialized IT expertise
- Different applications/ software
- Application support for Cloud technologies

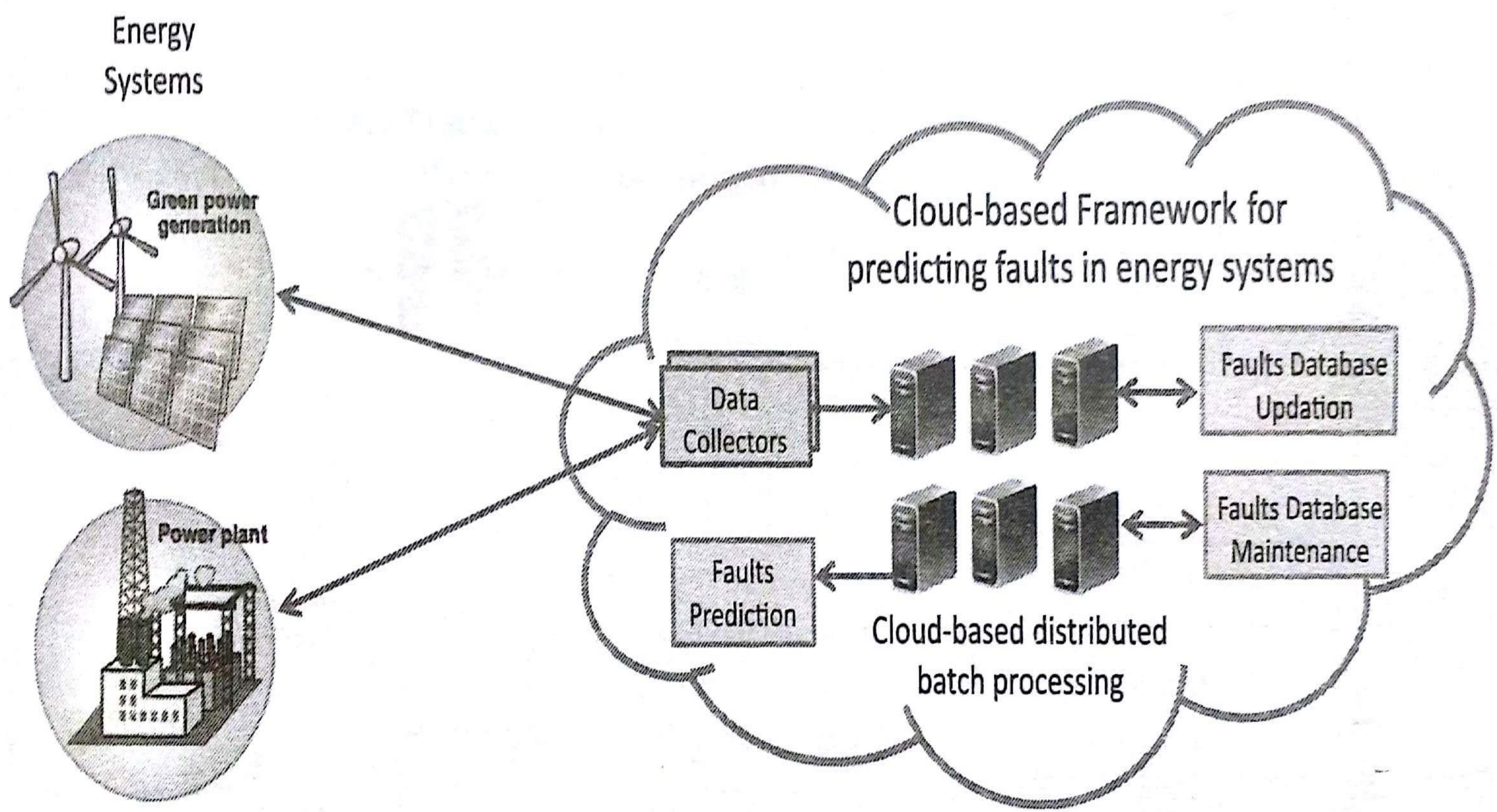
Cloud-based Services and Applications

1.Healthcare



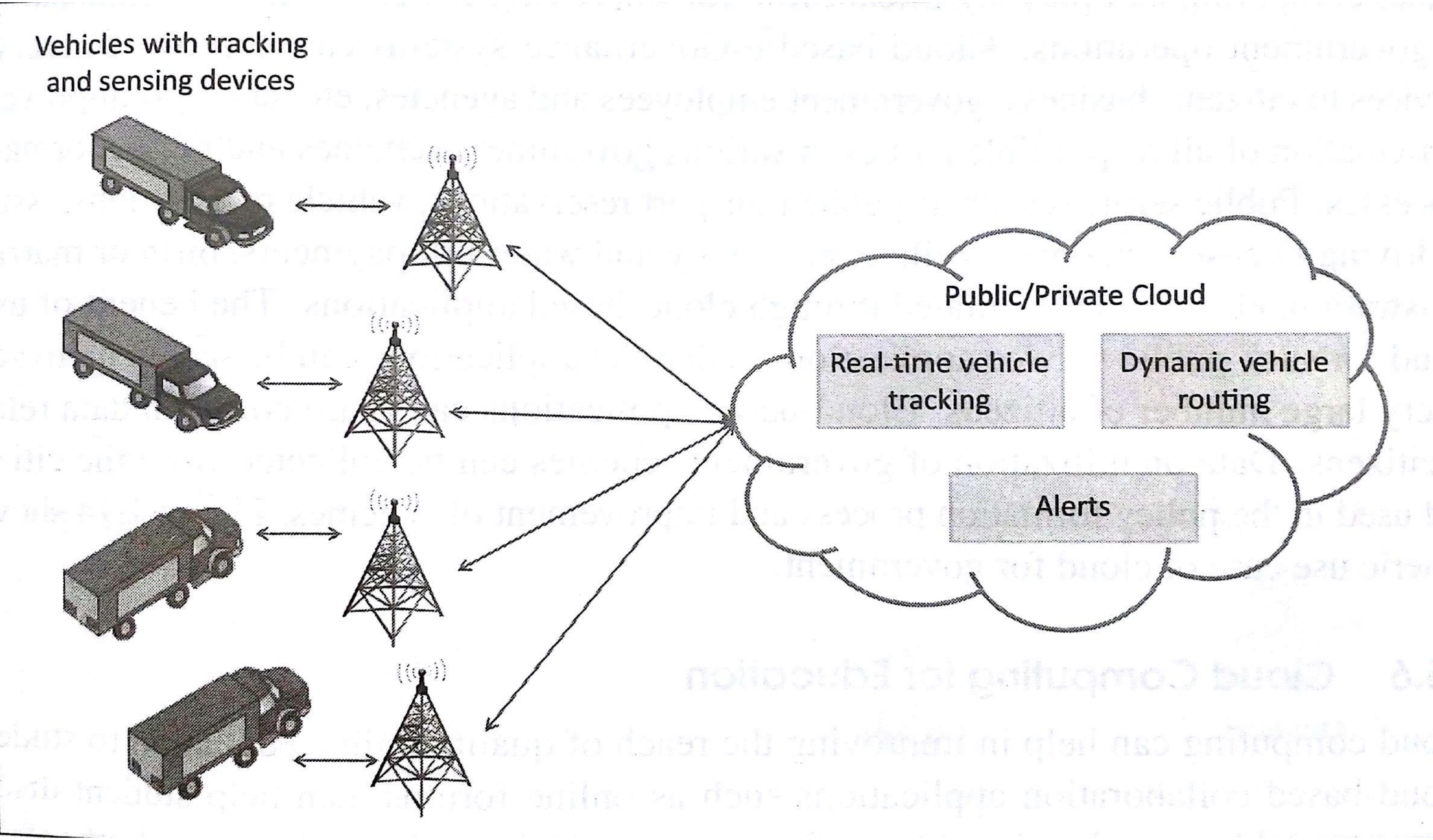
Cloud-based Services and Applications

2.Energy Systems



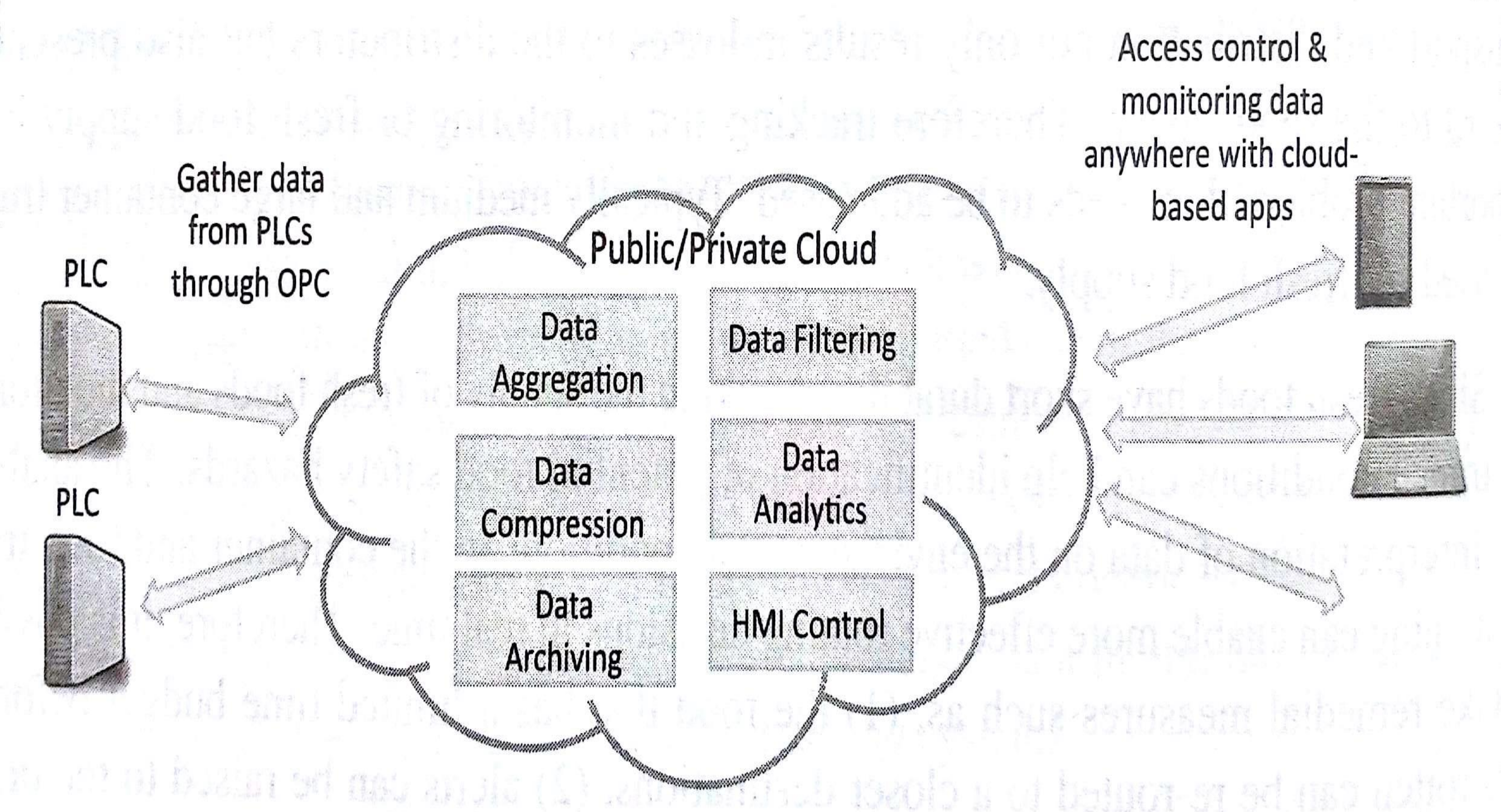
Cloud-based Services and Applications

3.Transportation System



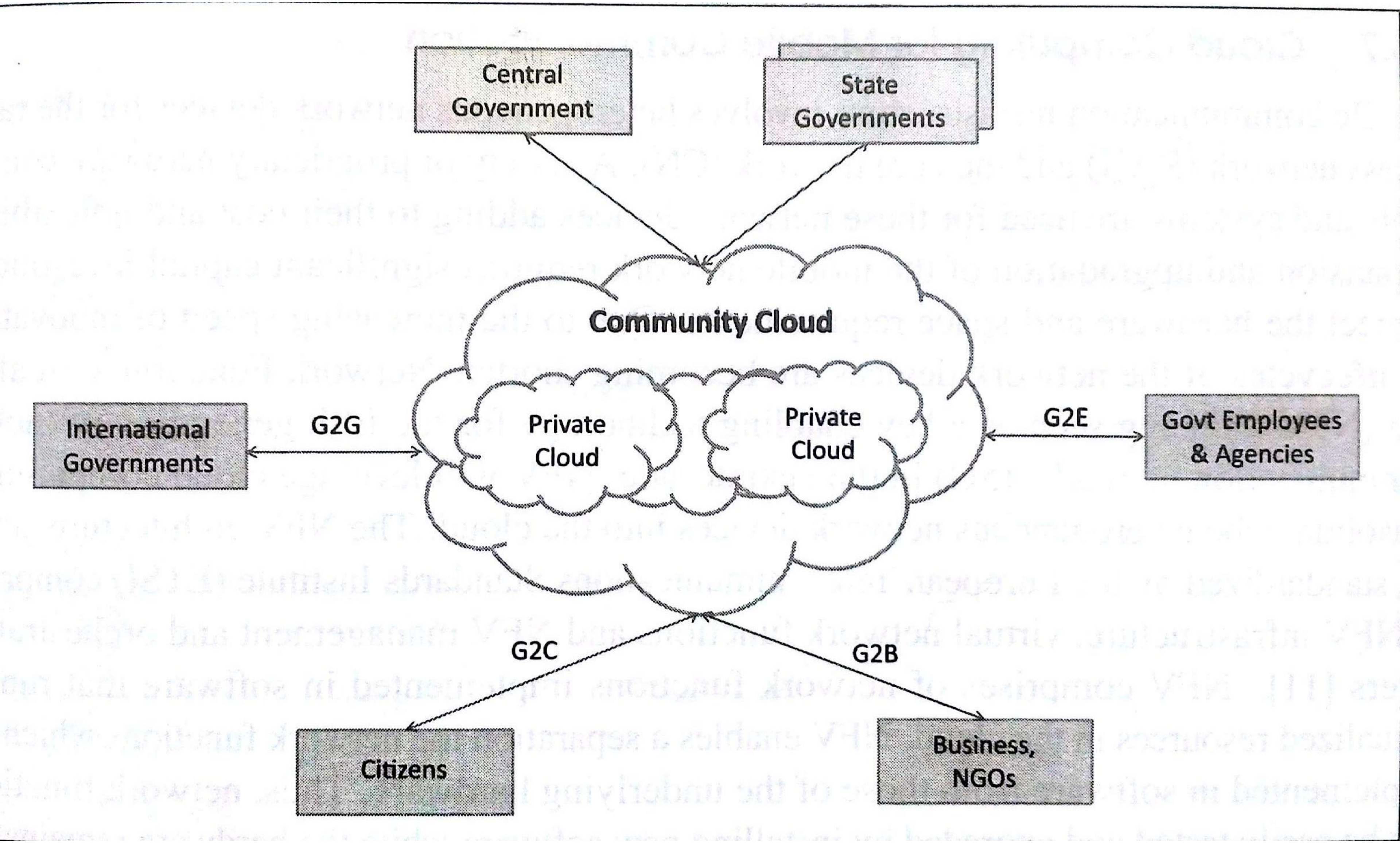
Cloud-based Services and Applications

4.Manufacturing Industry



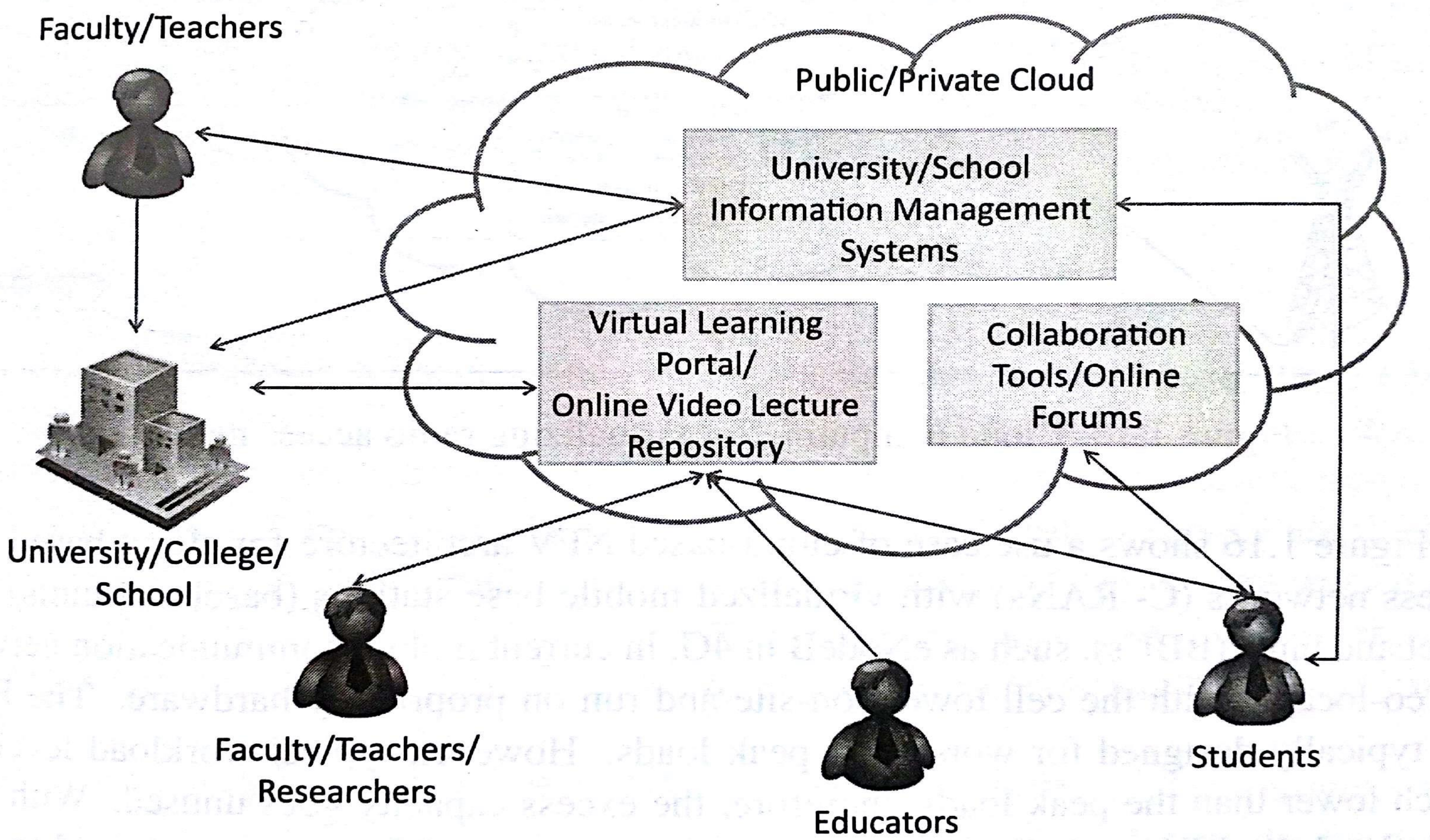
Cloud-based Services and Applications

5. Government



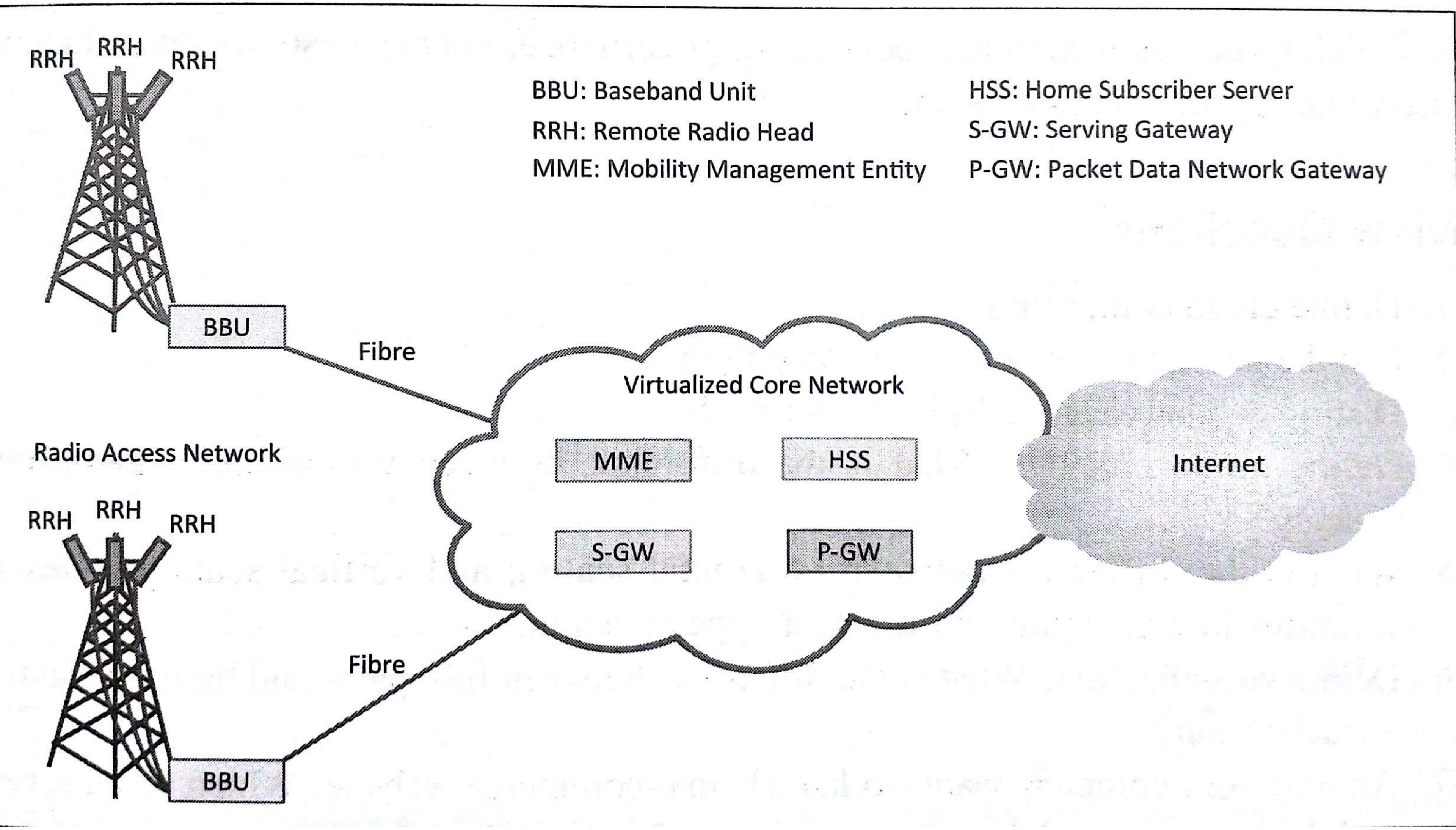
Cloud-based Services and Applications

6. Education



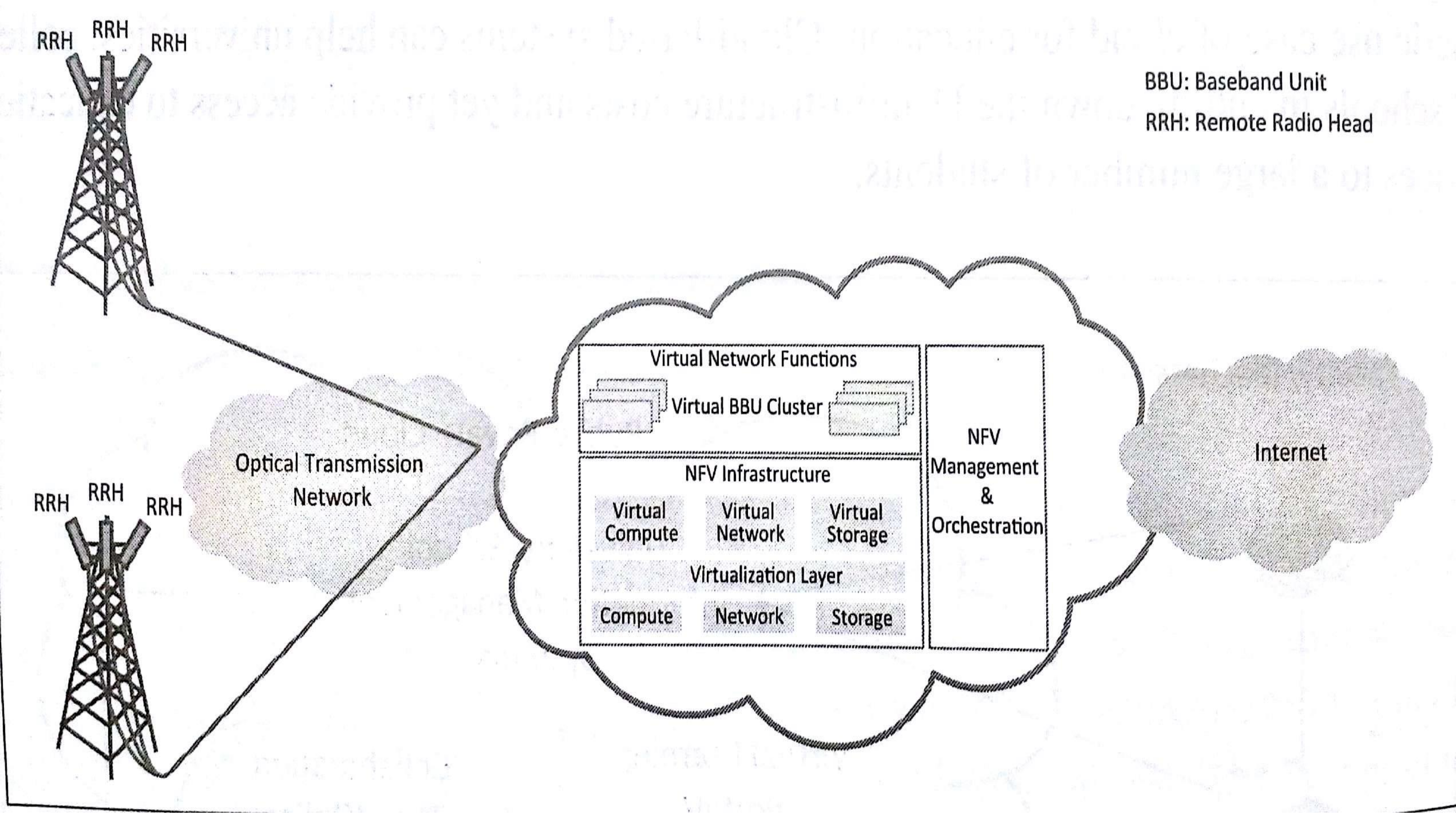
Cloud-based Services and Applications

7. a. Mobile Communication



Cloud-based Services and Applications

7.b. Mobile Communication



Thank You